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Final Report

Classification of Objects at a Very Low Resolution
Robustness of Deep Visual Models under THz-like Degradation

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Abstract

Background. Terahertz (THz) imaging sensors produce frames that are simultaneously low resolution, low signal-to-noise, and colorless. ImageNet-pretrained classifiers, which rely on fine textures and saturated chroma, lose accuracy quickly under those conditions, but the magnitude and the mechanism of the loss are not documented quantitatively in the literature.

Aim. Measure how three representative deep architectures (a residual CNN, a densely connected CNN, and a foveal vision transformer) degrade under controlled synthetic THz-like degradation, decompose the loss into per-axis contributions, and quantify how much of it can be recovered by training-time regularization.

Methods. The 402-cell campaign trains ResNet50, DenseNet121 and TransNeXt-tiny on CIFAR-10 and MNIST under a five-level degradation curve (saturation, downsample, blur, additive noise, salt and pepper). Seven phases isolate the effects: A clean baseline, B combined, C single-axis, D regularization recovery, B2 / B2-nr THz protocol simplification, and C2 single-axis under the THz protocol. A 48-replicate multi-seed audit ({42, 43, 44}) on the 24 L3 headline cells bounds reproducibility. Every degradation pipeline is anchored to a deterministic 256-image PSNR / SSIM probe so the cross-phase comparison sits on a physical scale.

Findings. On CIFAR-10 the combined L3 cell costs roughly 12.5 percentage points of accuracy relative to the clean baseline; regularization (T3 in Phase D) recovers ~ 0.77 percentage points of that drop. Resolution dominates the per-axis attribution by a factor of ~ 4 (mean L1 to L5 effect: -27.45 percentage points vs -6.58 for blur and -5.13 for salt and pepper). The multi-seed audit confirms $\sigma < 1.5$ percentage points on every L3 headline cell. Expected calibration error inflates from below 5 percentage points at L3 to above 15 percentage points at L5 on CIFAR-10: confidence is preserved even as the spatial signal collapses.

Conclusions. The Phase B accuracy collapse is best modeled as an information bottleneck, not as an overfit. The B2 / B2-nr decomposition shows that THz-protocol simplification recovers ~ 3.24 percentage points of pure-protocol headroom on top of the regularization recovery. Single-axis attribution implies that any THz deployment should invest in resolution restoration before denoising or deblurring.

Keywords. object classification, deep learning, low resolution, THz imaging, robustness, PSNR, SSIM, calibration, transformer, residual network, dense network.

Hebrew abstract. The Hebrew translation required by regulation §10.7.2.3 is supplied inside the DOCX, in Appendix A. Right-to-left text is rendered by Word natively; the LaTeX reference PDF reproduces only the English version.

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1 Introduction

1.1 Motivation

Terahertz (THz) imaging is increasingly used in security screening, biomedical diagnostics, and through-fabric inspection because the millimeter-wave band penetrates clothing, paper, plastics, and other low-permittivity materials that block visible light. The physics of the band exacts a price: diffraction caps the achievable spatial resolution near the wavelength of the source, single-pixel detectors integrate the field over a comparatively large area, and the available noise figure is dominated by thermal background. The output is an image that is simultaneously low resolution, low signal-to-noise, and free of color information.

Standard deep-vision classifiers, almost all of which inherit ImageNet pretraining, are tuned for the opposite distribution. They depend on fine textures, sharp edges, and saturated chroma to populate the receptive fields of the early convolutional layers. When those signals are removed the accuracy of an off-the-shelf model collapses, and the failure mode is opaque: it is not obvious how much of the loss is caused by reduced resolution alone, how much by additive noise, and how much by the loss of color, nor whether training-time interventions (regularization, augmentation, architectural substitution) can recover any of the lost performance.

1.2 Problem statement

This project quantifies the accuracy degradation of three representative deep architectures under a controlled synthetic THz-like degradation, decomposes the loss into per-axis contributions, audits the reproducibility of the measurements across random seeds, and documents which training-side interventions recover lost accuracy. The architectures span the three families that currently anchor the image-classification literature: a residual convolutional network (ResNet50 [1]), a densely connected convolutional network with explicit feature reuse (DenseNet121 [2]), and a vision transformer with foveal attention (TransNeXT-tiny [3]).

The success criteria taken from the preliminary report are: at least 80 percent top-1 accuracy on the moderate-degradation operating point, an identifiable architectural winner that beats the baseline by a statistically meaningful margin, and inference latency compatible with sensor-side deployment. Those targets are revisited in Section 6 against the actual measurements.

1.3 Alternative approaches considered

Three alternative families were reviewed during the preliminary design and rejected in favor of synthetic-degradation training.

(i) Denoise then classify. A pipeline that restores the image before invoking a clean-image classifier. The approach decouples the two problems and reuses an existing pretrained classifier verbatim, but it requires a high-quality reference image at train time, which a real THz sensor never produces. Restoration networks trained on mismatched degradation distributions tend to hallucinate detail [4, 5], which inflates PSNR / SSIM scores without improving downstream classification.

(ii) Super-resolution then classify. A specialization of (i) that targets the resolution axis only. It assumes that the dominant failure mode is spatial sampling. As Section 4 will show, resolution does dominate the per-axis attribution, but additive noise and blur still cost between five and seven percentage points each at the moderate operating point, so a resolution-only restoration cannot recover the full loss.

(iii) Train-time degradation exposure. The model is trained directly on synthetically degraded images so that the representation adapts to the degraded distribution. This is the approach used in modern robustness benchmarks [6] and the one adopted here. It is sample-efficient (no separate restoration network), it keeps the test-time pipeline minimal (single forward pass), and it surfaces the architectural sensitivities directly because every backbone sees the same degraded distribution.

1.4 Contributions

This work contributes:

- a 402-cell training campaign across three architectures, two datasets, five degradation levels, and six experimental phases, all trained to convergence under a single fair-comparison invariant (224×224 uniform input, frozen Optuna L3 hyperparameters, identical data pipeline);
- a 48-replicate multi-seed audit on the 24 L3 headline cells that bounds run-to-run variance below 1.5 percentage points;
- a deterministic 256-image PSNR / SSIM probe per pipeline so that accuracy can be compared against the physical image-quality scale, not only against the discrete level index;
- a per-axis attribution decomposition at L3 that shows resolution is the dominant cost ($\sim 4\times$ blur or salt and pepper);
- a regularization-recovery sweep (Phase D) and a protocol simplification sweep (Phase B2 / B2-nr) that, together, quantify the fraction of the accuracy collapse that is information-bottleneck versus training noise.

1.5 Roadmap

Section 2 reports the as-built technical specification of the delivered software product and lists every drift from the preliminary specification. Section 3 documents the engineering approach and the mathematical model. Section 4 reports the final acceptance tests, including the per-pipeline image-quality probe and the

seven per-phase accuracy tables. Section 5 enumerates the problems encountered during implementation and how each was resolved. Section 6 compares the achieved performance to the preliminary specification and discusses goal-by-goal success.

2 Final Technical Specification

This section reports the as-built specification of the delivered software product, and documents every drift from the preliminary specification [2, 3].

2.1 Product overview

The delivered product is a research-grade Python package that:

1. synthetically degrades a clean image dataset using a deterministic five-axis pipeline (saturation, downsample, blur, additive noise, salt and pepper);
2. fine-tunes one of three pretrained deep backbones (ResNet50, DenseNet121, TransNeXt-tiny) on the degraded distribution under a unified training recipe;
3. evaluates the trained model on a held-out validation split and records validation accuracy, calibration, inference throughput, and per-pipeline PSNR / SSIM metrics;
4. aggregates results across the 402-cell campaign and the 48-replicate multi-seed audit into a single canonical JSON artifact and a 7-tab interactive dashboard.

2.2 Inputs, outputs, interfaces

Inputs. Standard image datasets (CIFAR-10 [7] and MNIST [8]), automatically downloaded by torchvision on first invocation. A YAML / dictionary configuration object selects the backbone, the degradation phase, the level, the axis (for single-axis phases), the treatment (for Phase D), and the random seed.

Outputs. Per-cell training logs, validation metrics, best checkpoint, image-quality probe JSON. The canonical aggregator (`Final_Exp.json`, schema version 3) joins the 402 canonical cells with the 48 multi-seed replicates and exposes $\langle \text{val_acc} \rangle \pm \sigma$ pairs.

Hardware interface. The training stack runs on a single NVIDIA RTX 5070 (sm_120, Blackwell architecture) at bf16-mixed precision. A CPU-only fallback (32-true precision) is available for portability but is too slow to be useful at the campaign scale. There is no specialized hardware development; the system targets commodity GPU servers.

2.3 Final specification of the system

Models. Three backbones are supported. Table 1 summarizes the training recipe; the differential learning-rate convention (head $\times 10$ over backbone) is taken from the source papers.

Table 1: Training hyperparameters (final).

Parameter	CNNs (ResNet50, DenseNet121)	TransNeXt-tiny
Input resolution	224×224	224×224
Optimizer	AdamW [9]	AdamW
Schedule	Cosine annealing	Cosine annealing
Head learning rate	10^{-3}	5×10^{-4}
Backbone learning rate	10^{-4}	5×10^{-5}
Weight decay	10^{-4}	5×10^{-2}
Label smoothing	0.1	0.1
Stochastic depth	0.0	0.1
Gradient clip	$\ \nabla\ _2 \leq 1.0$	$\ \nabla\ _2 \leq 1.0$
Batch size	32	32
Max epochs	60	60
Early stopping	patience 10	patience 10
Precision	bf16-mixed	bf16-mixed
Seed	42 (canonical)	42 (canonical)

Degradation pipeline. The five-level schedule (Table 2) is the single source of truth for L1 through L5; it is imported from `src/data/degradation_levels.py` into both the training and the evaluation paths.

Table 2: Five-level synthetic degradation schedule. Blur kernel and σ are pixel-domain values at the 224×224 model input.

Level	Name	low_res	Blur K	Blur σ	Noise std	S&P	Saturation
L1	Mild	18	13	2.50	0.04	0.03	0.95
L2	Light	12	25	5.00	0.08	0.06	0.65
L3	Moderate	8	41	8.00	0.12	0.10	0.40
L4	Severe	6	61	12.00	0.16	0.14	0.15
L5	Extreme	3	91	18.00	0.22	0.18	0.00

Data pipeline. Pipeline version 2 (US-017, finalized 2026-05-20): the saturation lerp is applied first, then the bilinear downsample to `low_res`, then additive Gaussian noise and salt and pepper at the low resolution, then bicubic upsample to 224×224 , then Gaussian blur (kernel calibrated for 224×224), then a final clip to $[0, 1]$ and ImageNet normalization.

Determinism. The per-sample degradation seed is `idx + 107` on the validation split and `idx` on the training split. The byte-identity gate in `src/tests/test_degradation_determinism.py` asserts that two independent reads of the same validation index yield bit-identical pixels across freshly built datasets;

this is also verified across seeds 42, 43, and 44 by the multi-seed audit test.

2.4 Drift from the preliminary specification

The Preliminary Design Review (PDR, Pre-2026-061.pdf) defined the project scope. Five drifts occurred during implementation:

1. **TransNeXt variant: small to tiny.** The preliminary spec proposed `transnext_small` ($\sim 50\text{M}$ parameters); US-016 (2026-05-14) substituted `transnext_tiny` ($\sim 28\text{M}$ parameters) to keep the parameter count comparable to ResNet50 ($\sim 25\text{M}$) and to recover ~ 310 GPU-hours across the TransNeXt rows. The change preserves the foveal-attention mechanism, which was the architectural property of interest, and it matches the published TransNeXt 224×224 checkpoint exactly.
2. **Uniform 224×224 protocol.** A short-lived V3 refactor (2026-05-08 through 2026-05-13) attempted native-resolution training (32×32 for TransNeXt on CIFAR-10, $28 \rightarrow 32$ for MNIST). The refactor violated the fair-comparison invariant (different backbones saw different tensor shapes), so it was rolled back. Every cell now trains at 224×224 .
3. **Blur kernel rescaling.** The 2026-05-12 degradation curve used blur kernels designed for native 32×32 imagery ($K = 3 \dots 13$, $\sigma = 0.8 \dots 2.3$). At 224×224 those kernels covered ~ 1 to 6% of the image width and produced essentially invisible blur. They were rescaled on 2026-05-14 to pixel-domain values at 224×224 ($K = 13 \dots 91$, $\sigma = 2.5 \dots 18$), with the kernel-to- σ rule $K = 2\lceil 2.5\sigma \rceil + 1$ so the kernel always captures the 2.5σ Gaussian footprint.
4. **Optuna freeze.** The preliminary plan reserved a per-(phase, cell) hyperparameter sweep. The final spec freezes the Optuna L3 winner per (model, dataset) and reuses it across every level and every axis. The change saves ~ 200 GPU-hours and removes a per-phase confound from the cross-phase comparison.
5. **Multi-seed audit.** The preliminary spec used a single seed (42). US-042 (2026-06-01) added a 48-replicate audit at seeds 43 and 44 across the 24 L3 headline cells so that the published L3 accuracies carry an explicit σ bound rather than a single point estimate.

2.5 Phases of the experimental program

The 402-cell campaign is partitioned into seven phases. Table 3 summarizes each phase.

3 Engineering Design

This section documents the chosen approach in detail. The system block diagram is shown in Figure 1; the degradation pipeline that drives the experimental program is shown in Figure 2.

Table 3: Experimental phases of the 402-cell campaign. Multi-seed audit replicates are layered on top of the 24 L3 headline cells.

Phase	Purpose	Cells	Tag prefix
A	Clean baseline (no degradation)	6	final_clean_
B	Combined degradation, all axes at level L	30	final_B_L*
C	Single-axis isolation, others at identity	150	final_C_L*_<axis>_
D	Regularization recovery (T1 / T2 / T3) on Phase B	90	final_D_T*_L*_
B2	THz protocol simplification (sat = 0, noise = 0, T3)	30	final_B2_L*_
B2nr	B2 without regularization, L3 only	6	final_B2nr_L3_
C2	Single-axis under THz protocol	90	final_C2_L*_<axis>_
Audit	Multi-seed audit (seeds {43, 44}) on the 24 L3 headline cells	48	*_seed{43, 44}
Total		450	

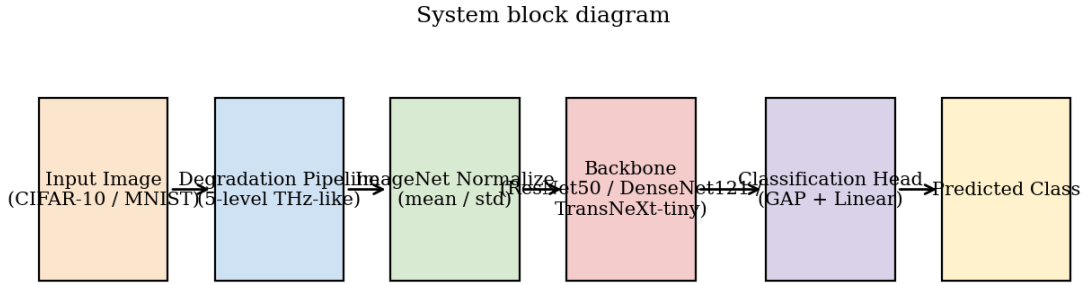


Figure 1: System block diagram. An input image from CIFAR-10 or MNIST passes through the THz-like degradation pipeline, is normalized to the ImageNet distribution, is fed to one of the three backbones, and is classified by a global-average-pooling head.

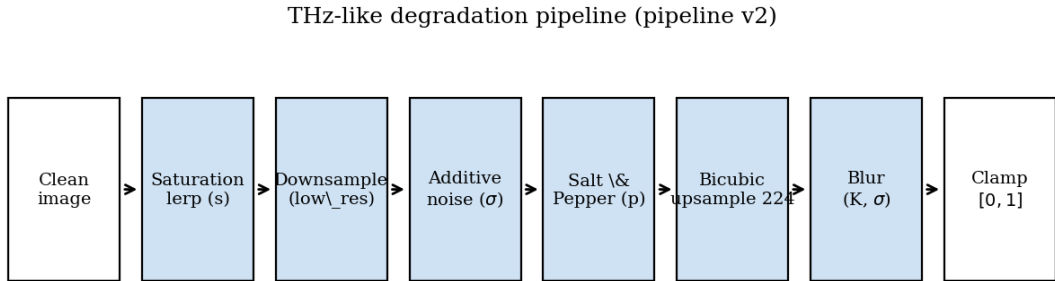


Figure 2: Pipeline version 2 (US-017). The saturation lerp runs first so the grayscale luminance is computed from the un-degraded source. Additive noise and salt-and-pepper operate at the pre-upsample resolution; bicubic upsample spreads each stochastic sample over the kernel footprint, producing sensor-grain noise structure rather than fine pixel-level grain.

3.1 Mathematical model

Degradation operator. Let $x \in [0, 1]^{3 \times H \times W}$ be a clean image and let $\phi = (s, r, \sigma_n, K_b, \sigma_b, p)$ denote the degradation parameters: saturation $s \in [0, 1]$, low-resolution target $r \in \mathbb{N}$, noise standard deviation $\sigma_n \geq 0$, blur kernel size K_b , blur sigma σ_b , and salt-and-pepper amount $p \in [0, 1]$. Define the composite operator

$$\mathcal{D}_\phi(x) = \text{Clip}_{[0,1]} \circ \text{Blur}_{K_b, \sigma_b} \circ \text{Up}_{224} \circ \text{SP}_p \circ \text{N}_{\sigma_n} \circ \text{Down}_r \circ \text{Sat}_s, \quad (1)$$

where $\text{Sat}_s(x) = (1 - s) \text{Gray}(x) + s x$ with $\text{Gray}(x) = 0.2989 x_R + 0.5870 x_G + 0.1140 x_B$ (BT.601 luma); Down_r is bilinear downsample to $r \times r$; N_{σ_n} adds independent Gaussian noise of standard deviation σ_n ; SP_p flips a $p/2$ fraction of pixels to 1 and another $p/2$ fraction to 0; Up_{224} is bicubic upsample to 224×224 ; and $\text{Blur}_{K_b, \sigma_b}$ is a separable Gaussian blur with kernel size K_b and standard deviation σ_b .

Classifier. A backbone $f_\theta(\cdot)$ produces a logit vector $z = f_\theta(\mathcal{D}_\phi(x))$. With label smoothing $\alpha = 0.1$, the training objective is the smoothed cross-entropy

$$\mathcal{L}(\theta) = - \sum_{c=1}^C \tilde{y}_c \log \text{softmax}(z)_c, \quad \tilde{y}_c = (1 - \alpha) \mathbf{1}[c = y] + \frac{\alpha}{C}. \quad (2)$$

Backbones. The three architectures expose different inductive biases for the low-resolution regime. ResNet50 [1] combines features via additive identity skips:

$$y_\ell = F_\ell(x_\ell; \theta_\ell) + x_\ell. \quad (3)$$

The summation reuses previous-layer features but collapses them into the running representation; subtle low-level cues can be washed out when the later residual blocks dominate.

DenseNet121 [2] concatenates the feature maps of every preceding layer in the same dense block:

$$x_\ell = H_\ell([x_0, x_1, \dots, x_{\ell-1}]), \quad (4)$$

where H_ℓ is a BatchNorm-ReLU-Conv_{3×3} composite. The collective knowledge of every preceding layer is explicitly preserved, which is the property the preliminary spec hypothesized would help under information loss.

TransNeXt-tiny [3] uses pixel-focused attention with foveal and peripheral kernels:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M_{\text{focal}}\right) V, \quad (5)$$

where M_{focal} is a window-shaped mask that down-weights all but the nearest neighbors of each query, while a parallel global pooling stream contributes a coarse-grained peripheral signal. The biomimetic

prior is designed to retain accuracy when the global shape is corrupted but local texture is partially preserved.

3.2 Image-quality metrics

Each degradation pipeline is anchored to a PSNR and SSIM probe so that cross-phase comparisons sit on a physical scale.

PSNR [4] between a reference image x and a degraded image $\tilde{x} = \mathcal{D}_\phi(x)$, both in $[0, 1]^{3 \times 224 \times 224}$, is

$$\text{PSNR}(x, \tilde{x}) = 10 \log_{10} \frac{1}{\text{MSE}(x, \tilde{x})}, \quad \text{MSE}(x, \tilde{x}) = \frac{1}{3 \cdot 224^2} \sum_{c,i,j} (x_{c,i,j} - \tilde{x}_{c,i,j})^2. \quad (6)$$

SSIM [5] is the structural similarity index. For two local patches u, v with means μ_u, μ_v , variances σ_u^2, σ_v^2 , and covariance σ_{uv} ,

$$\text{SSIM}(u, v) = \frac{(2\mu_u\mu_v + C_1)(2\sigma_{uv} + C_2)}{(\mu_u^2 + \mu_v^2 + C_1)(\sigma_u^2 + \sigma_v^2 + C_2)}, \quad (7)$$

with C_1, C_2 small stabilizers; the full-image score is the spatial mean over an 11×11 Gaussian window. The implementation uses `torchmetrics.image.StructuralSimilarityIndexMeasure` with `data_range = 1`.

Reference construction. The PSNR / SSIM reference is the input image upsampled to 224×224 by bilinear interpolation. The choice matches what a hypothetical clean-image classifier would see if the degradation pipeline were short-circuited; it does not include the saturation lerp, so the measurements isolate the spatial loss from the color loss. Both choices are documented in the figure captions in Section 4 so the reader can interpret the absolute scale correctly.

Probe protocol. For each pipeline the probe iterates over the first 256 validation indices, applies the pipeline with seed offset 2×10^7 (chosen disjoint from the training and validation streams), and aggregates the mean and standard deviation. The 256-sample size balances probe runtime against the standard error of the mean (below 0.1 dB on the L3 pipelines).

3.3 Calibration metric

Expected calibration error (ECE) [10, 11] bins the predicted probabilities into $M = 15$ equal-width bins and computes

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (8)$$

where B_m is the set of predictions whose top-1 probability falls in bin m , $\text{acc}(B_m)$ is the empirical accuracy of B_m , and $\text{conf}(B_m)$ is the mean predicted probability in B_m . A well-calibrated model has

$ECE \rightarrow 0$.

3.4 Regularization treatments (Phase D)

Phase D layers three regularization treatments on top of the frozen Phase B Optuna winners:

T1. Dropout (CNN, rate 0.2) or stochastic-depth drop-path [12] (TransNeXt, rate 0.2). Architectural regularization only; no label-side intervention.

T2. mixup [13] with $\alpha = 0.2$; label-mixing only.

T3. The combination: dropout (or drop-path) + mixup $\alpha = 0.2$ + CutMix [14] $\alpha = 1.0$. The kitchen-sink combination defines the upper bound of training-side recovery measurable inside the design space.

3.5 Software architecture

The implementation is layered:

- `src/data/` hosts the single source of truth for the degradation curve (`degradation_levels.py`) and the operator (`degrade.py`). Both modules are imported verbatim into the training and the image-quality probe paths, which is the property that guarantees the measured PSNR / SSIM correspond to the pixels the model actually sees.
- `src/lightning/` hosts the training engine (`THzClassifier`, `THzDataModule`) built on PyTorch Lightning [15].
- `src/models/` hosts the backbone wrappers; the TransNeXt weights are auto-downloaded from the public checkpoint and cached under `artifacts/weights/`.
- `scripts/` hosts the campaign drivers (`run_all_phases.py`, `run_ralph_loop.py`) and the diagnostics builders.
- `Final_Report/scripts/` hosts the figure / table / sample generators that produce this report. Every artifact under `Final_Report/` is reproducible by running `python Final_Report/scripts/build_*.py` from a clean checkout.

4 Final Acceptance Tests

This section reports the measured performance of the delivered product against every relevant metric. Every value comes from a frozen training run recorded under `runs/final/` and aggregated by `artifacts/Final_Exp.json` (schema version 3). No values are synthesized or interpolated.

4.1 Test rig

The campaign ran on a single workstation with the configuration in Table 4. Determinism is guarded by `pl.seed_everything(42, workers=True)` for the canonical seed and by the per-sample seed offset for the validation split.

Table 4: Test rig and software stack.

Component	Value
GPU	NVIDIA RTX 5070 (Blackwell, sm_120, 12 GB)
Precision	bf16-mixed
CPU dataloader workers	0 (Windows)
Framework	PyTorch 2.11 [16] + Lightning 2.x [15]
Image quality	torchmetrics PSNR / SSIM, data_range = 1
Determinism gate	src/tests/test_degradation_determinism.py

4.2 Per-pipeline image quality

Table 5 reports the mean PSNR and SSIM for every degradation pipeline that appears in the experiments. The values are identical across same-pipeline cells by construction, so Phase D rows inherit the Phase B PSNR / SSIM at the same level. Figures 3 and 4 show the visual appearance of the pipelines on CIFAR-10; Figures 5 and 6 repeat the comparison on MNIST.

4.3 Clean baseline accuracy (Phase A)

Table 6 reports the validation accuracy at the clean operating point. These values are the upper bounds against which every subsequent phase is measured.

4.4 Combined-degradation accuracy (Phase B)

Phase B applies every axis simultaneously at the same level. Table 7 reports the full $3 \times 2 \times 5$ accuracy block; Figure 7 plots the curves with the Phase D-T3 recovery overlay.

4.5 Accuracy versus image quality

Figures 8 and 9 plot validation accuracy against the per-pipeline mean PSNR and SSIM respectively. The Spearman rank correlation is reported in each panel and is monotonically strong on both metrics for both datasets. Accuracy and image quality collapse together, but the slope is not unity: on CIFAR-10 the model loses accuracy faster than SSIM, while on MNIST the two metrics track each other closely. This corroborates the per-axis attribution in Section 4.6: CIFAR-10 carries more inter-class color information that is destroyed by the saturation lerp, and SSIM does not penalize the color collapse strongly.

Table 5: Mean PSNR and SSIM per degradation pipeline. Each value is computed over a fixed deterministic 256-sample validation subset using torchmetrics with data_range=1, sample seed offset = 2×10^7 , clean reference upsampled to 224×224 by bilinear interpolation.

Dataset	Phase	Level	Axis	PSNR (dB)	SSIM
CIFAR-10	B	L1	–	21.23 ± 1.54	0.801 ± 0.041
CIFAR-10	B	L2	–	17.54 ± 1.32	0.697 ± 0.059
CIFAR-10	B	L3	–	14.87 ± 1.41	0.635 ± 0.078
CIFAR-10	B	L4	–	13.10 ± 1.41	0.601 ± 0.087
CIFAR-10	B	L5	–	10.85 ± 1.81	0.566 ± 0.110
CIFAR-10	B2	L1	–	19.50 ± 2.31	0.816 ± 0.054
CIFAR-10	B2	L2	–	17.25 ± 1.88	0.733 ± 0.067
CIFAR-10	B2	L3	–	15.24 ± 1.72	0.673 ± 0.079
CIFAR-10	B2	L4	–	14.18 ± 1.86	0.652 ± 0.086
CIFAR-10	B2	L5	–	12.40 ± 2.50	0.624 ± 0.108
CIFAR-10	B2nr	L3	–	15.24 ± 1.72	0.673 ± 0.079
CIFAR-10	C	L1	Blur	41.98 ± 2.54	0.983 ± 0.007
CIFAR-10	C	L1	Noise	28.60 ± 0.42	0.841 ± 0.021
CIFAR-10	C	L1	Resolution	30.78 ± 2.70	0.917 ± 0.035
CIFAR-10	C	L1	S&P	21.66 ± 1.10	0.858 ± 0.023
CIFAR-10	C	L1	Saturation	36.45 ± 2.39	0.975 ± 0.010
CIFAR-10	C	L2	Blur	33.58 ± 2.59	0.952 ± 0.020
CIFAR-10	C	L2	Noise	23.34 ± 0.36	0.642 ± 0.038
CIFAR-10	C	L2	Resolution	25.33 ± 2.65	0.845 ± 0.060
CIFAR-10	C	L2	S&P	18.61 ± 0.87	0.755 ± 0.032
CIFAR-10	C	L2	Saturation	31.34 ± 3.42	0.968 ± 0.014
CIFAR-10	C	L3	Blur	27.22 ± 2.54	0.885 ± 0.045
CIFAR-10	C	L3	Noise	20.12 ± 0.42	0.489 ± 0.047
CIFAR-10	C	L3	Resolution	21.75 ± 2.64	0.784 ± 0.079
CIFAR-10	C	L3	S&P	16.38 ± 0.70	0.635 ± 0.035
CIFAR-10	C	L3	Saturation	27.94 ± 4.23	0.958 ± 0.022
CIFAR-10	C	L4	Blur	23.63 ± 2.52	0.822 ± 0.066
CIFAR-10	C	L4	Noise	17.87 ± 0.44	0.381 ± 0.050
CIFAR-10	C	L4	Resolution	19.42 ± 2.67	0.745 ± 0.091
CIFAR-10	C	L4	S&P	14.96 ± 0.60	0.538 ± 0.038
CIFAR-10	C	L4	Saturation	25.41 ± 4.72	0.945 ± 0.033
CIFAR-10	C	L5	Blur	21.06 ± 2.52	0.772 ± 0.082
CIFAR-10	C	L5	Noise	15.49 ± 0.43	0.277 ± 0.048
CIFAR-10	C	L5	Resolution	15.88 ± 2.82	0.692 ± 0.109
CIFAR-10	C	L5	S&P	13.88 ± 0.58	0.456 ± 0.041
CIFAR-10	C	L5	Saturation	24.17 ± 4.93	0.936 ± 0.041
CIFAR-10	C2	L1	Blur	24.67 ± 5.66	0.943 ± 0.040
CIFAR-10	C2	L1	Resolution	23.11 ± 3.97	0.883 ± 0.050
CIFAR-10	C2	L1	S&P	19.27 ± 2.05	0.824 ± 0.043
CIFAR-10	C2	L2	Blur	23.74 ± 4.42	0.915 ± 0.044
CIFAR-10	C2	L2	Resolution	21.22 ± 3.09	0.819 ± 0.068
CIFAR-10	C2	L2	S&P	17.19 ± 1.43	0.724 ± 0.044
CIFAR-10	C2	L3	Blur	22.05 ± 3.35	0.855 ± 0.057
CIFAR-10	C2	L3	Resolution	19.34 ± 2.65	0.763 ± 0.082
CIFAR-10	C2	L3	S&P ¹¹	15.46 ± 1.03	0.609 ± 0.044
CIFAR-10	C2	L4	Blur	20.43 ± 2.77	0.798 ± 0.071
CIFAR-10	C2	L4	Resolution	17.79 ± 2.47	0.729 ± 0.092

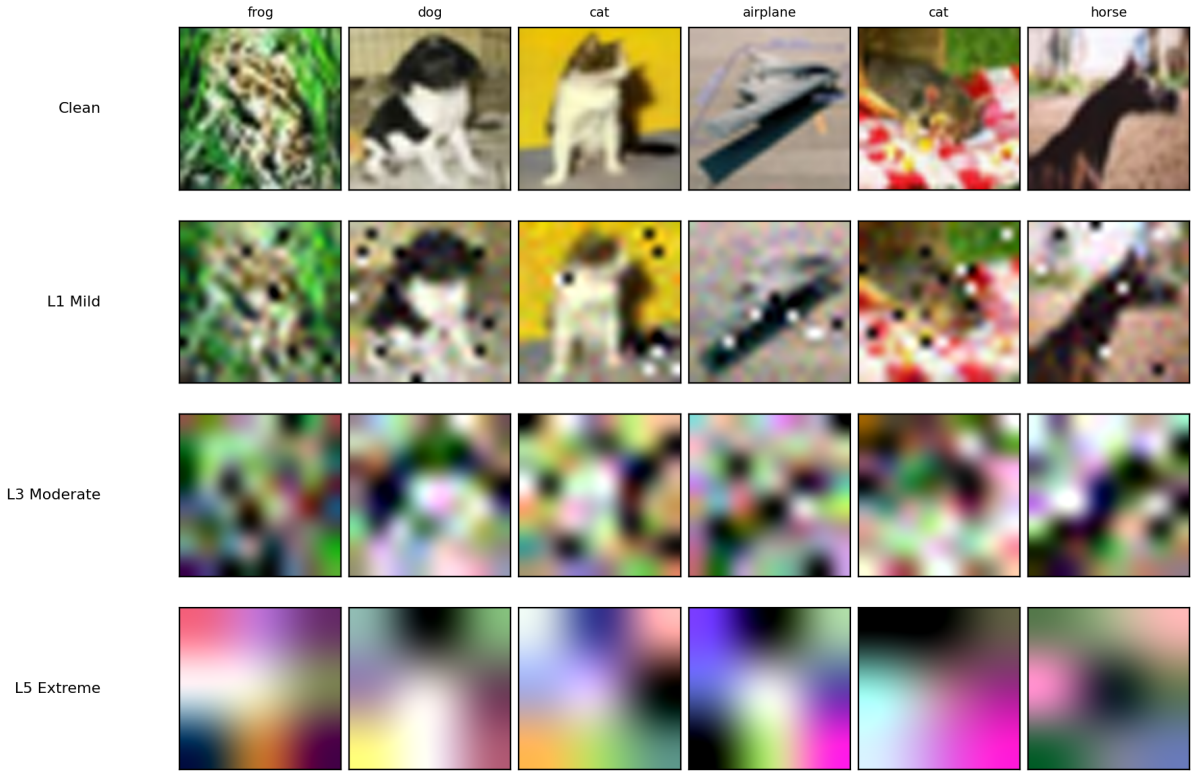


Figure 3: CIFAR-10: six fixed validation examples (deterministic seed) at clean, L1 Mild, L3 Moderate, and L5 Extreme combined degradation. Class labels above each column. Rows from top to bottom: Clean, L1, L3, L5.

Table 6: Phase A clean-baseline accuracy (%) on the held-out validation split. No degradation applied; values are upper bounds for every subsequent phase.

Model	CIFAR-10	MNIST
ResNet50	95.18	99.14
DenseNet121	93.56	99.30
TransNeXt-tiny	97.64	99.24

Table 7: Phase B (combined degradation) validation accuracy (%). All five axes active at the same level L. Cells with $\pm$ values are multi-seed audited (seeds {42, 43, 44}).

Model	Dataset	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	76.46	53.94	36.31 ± 0.83	26.84	19.06
ResNet50	MNIST	99.18	96.36	79.17 ± 0.80	56.44	28.00
DenseNet121	CIFAR-10	78.48	53.68	38.54 ± 0.78	28.52	19.58
DenseNet121	MNIST	98.74	96.54	80.79 ± 0.84	58.08	27.76
TransNeXt-tiny	CIFAR-10	87.58	64.22	39.48 ± 0.62	28.50	20.10
TransNeXt-tiny	MNIST	98.96	96.00	80.05 ± 0.11	57.24	27.48

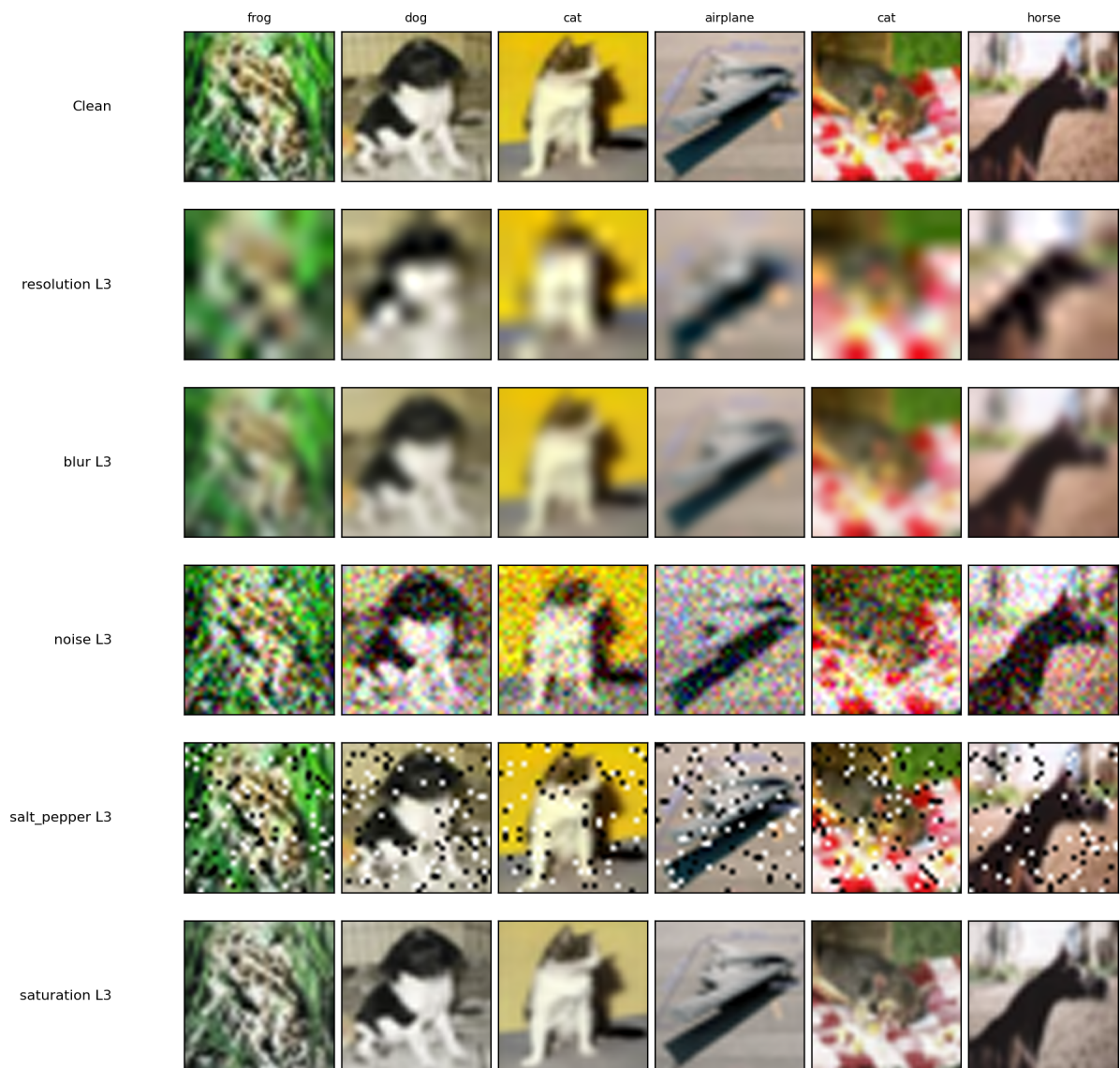


Figure 4: CIFAR-10: per-axis isolation at L3 (Phase C). Each row shows the same six images with only the named axis active.

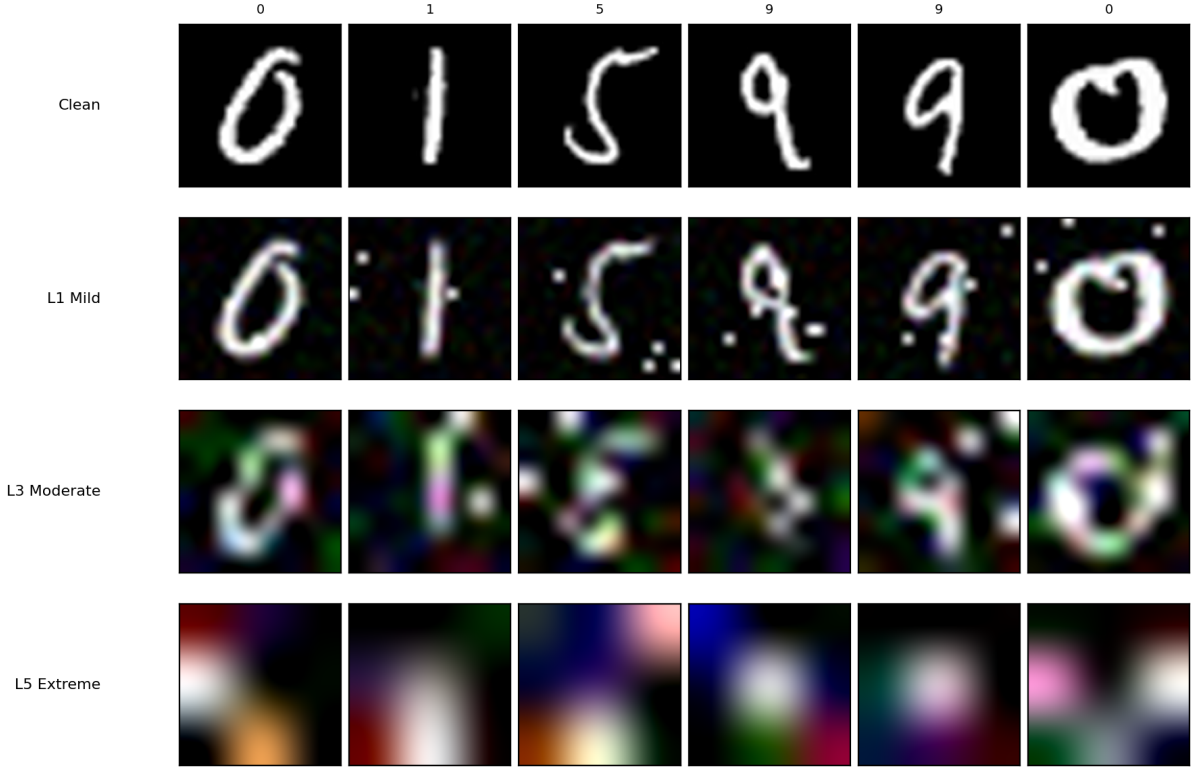


Figure 5: MNIST: combined degradation at L1, L3, L5 on six fixed validation examples.

4.6 Per-axis isolation (Phase C)

Phase C isolates each axis: the named axis is at level L , every other axis is at its identity value. Table 8 reports the full $3 \times 2 \times 5 \times 5 = 150$ accuracy block; Figure 10 shows the per-axis attribution at L3 as accuracy delta vs the Phase A clean baseline.

Table 9 reports the L3 attribution numerically together with the combined Phase B L3 cell so the reader can verify that the per-axis drops sum to less than the combined drop (the axes are not additive: their interactions compound).

4.7 Regularization recovery (Phase D)

Phase D layers three regularization treatments on top of the Phase B Optuna winners. Table 10 reports the accuracy at every level for every treatment.

4.8 THz protocol simplification (Phase B2 and B2-nr)

Phase B2 simplifies the THz protocol by forcing saturation = 0 (full grayscale) and noise = 0 (no additive noise). T3 regularization is still applied. Phase B2-nr is the no-regularization arm at L3 only, used to decompose the Phase B1 to B2 improvement into pure-protocol and regularization fractions. Table 11 reports the accuracy block.

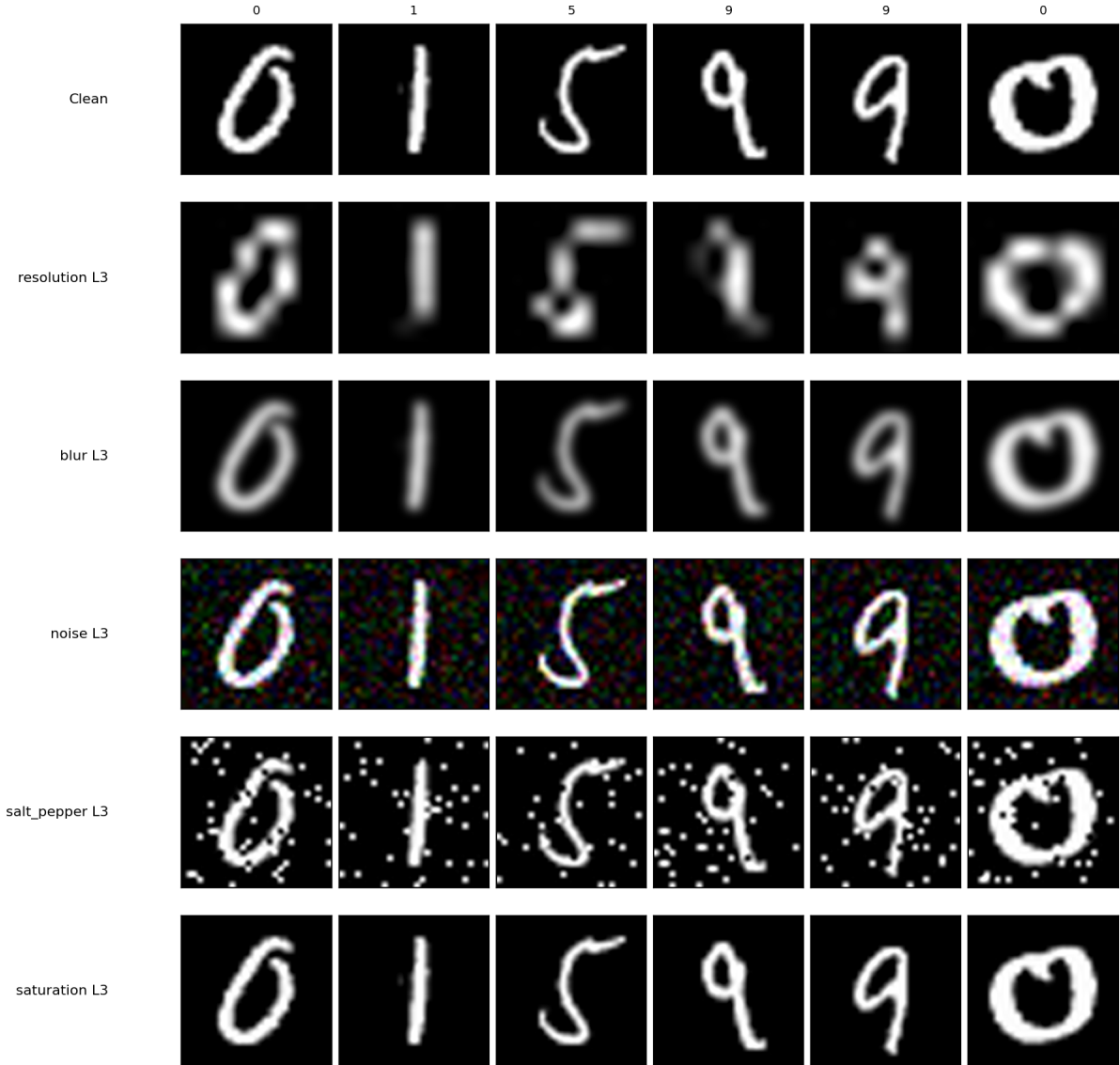


Figure 6: MNIST: per-axis isolation at L3 (Phase C).

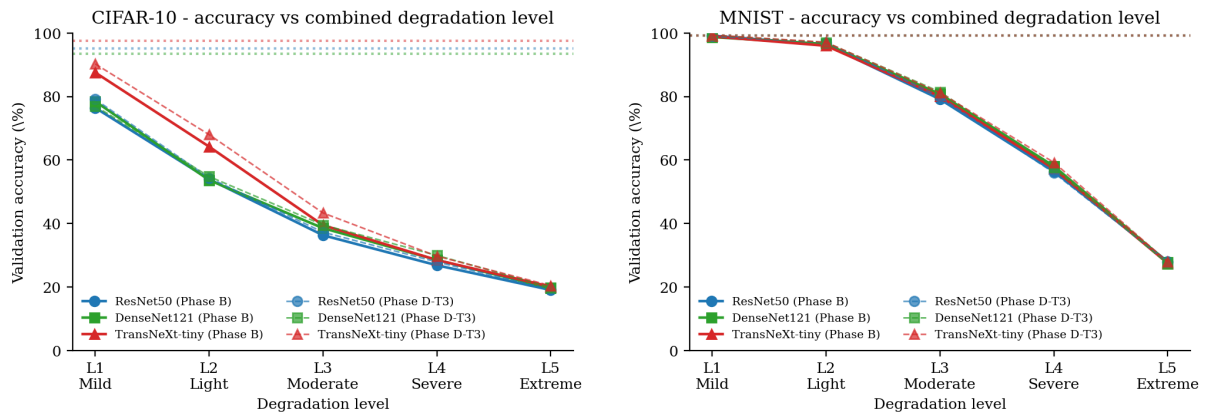


Figure 7: Validation accuracy vs combined degradation level. Solid curves: Phase B (no extra regularization). Dashed curves: Phase D-T3 (T3 regularization on the same pipeline). Dotted horizontal lines: Phase A clean baseline per model.

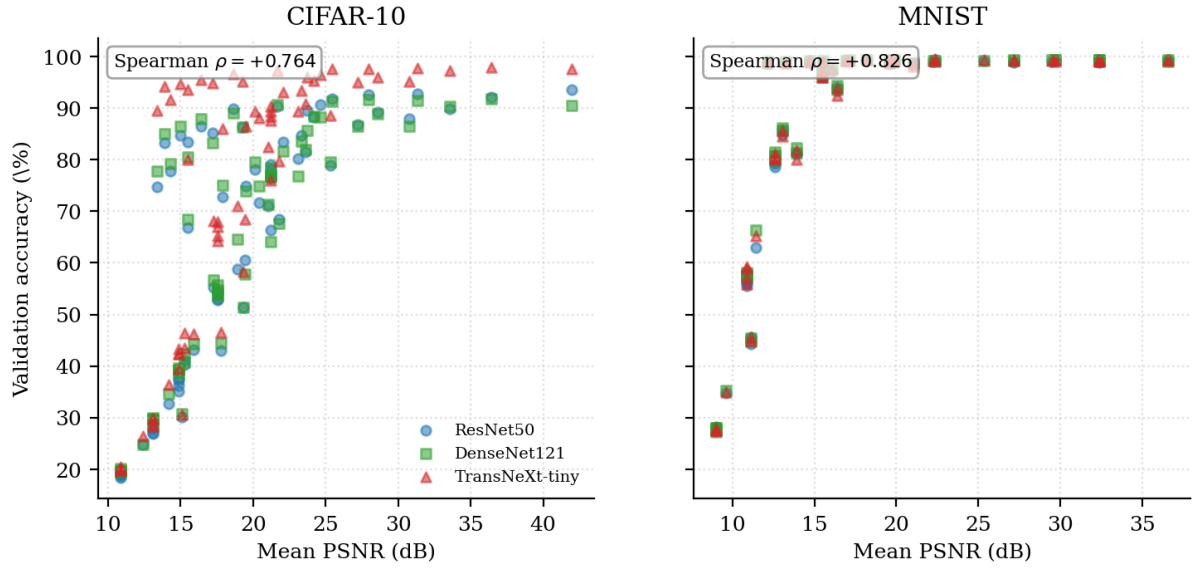


Figure 8: Validation accuracy vs mean PSNR per pipeline. Each marker is one trained cell; color indicates the backbone. The strong negative monotone relationship justifies treating PSNR as a low-cost proxy for downstream accuracy at design time.

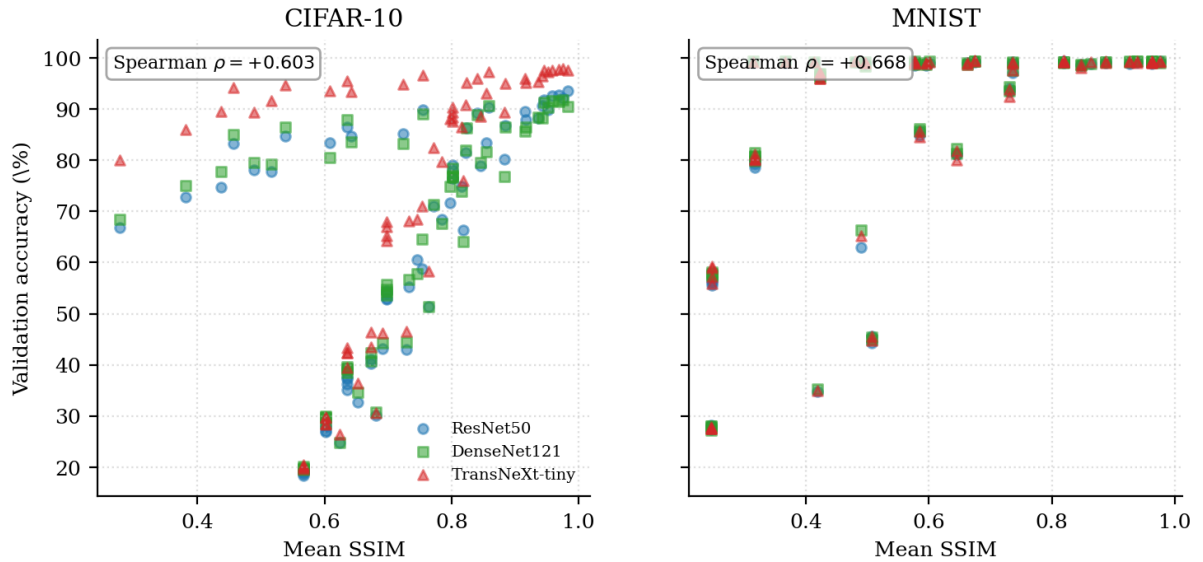


Figure 9: Validation accuracy vs mean SSIM per pipeline.

Table 8: Phase C single-axis isolation validation accuracy (%). The named axis is at level L; every other axis is at its identity value.

Model	Dataset	Axis	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	Resolution	88.00	78.98	68.46	60.50	43.16
ResNet50	CIFAR-10	Blur	93.54	89.88	86.78	81.46	70.98
ResNet50	CIFAR-10	Noise	89.14	84.70	78.06	72.76	66.88
ResNet50	CIFAR-10	S&P	90.42	89.92	86.52	84.64	83.30
ResNet50	CIFAR-10	Saturation	92.06	92.82	92.56	91.78	88.22
ResNet50	MNIST	Resolution	98.84	98.50	93.48	81.22	44.26
ResNet50	MNIST	Blur	99.32	99.14	99.18	99.20	98.82
ResNet50	MNIST	Noise	99.10	99.16	99.10	99.10	98.96
ResNet50	MNIST	S&P	99.08	98.98	98.88	98.90	98.62
ResNet50	MNIST	Saturation	98.94	99.18	99.36	99.18	99.10
DenseNet121	CIFAR-10	Resolution	86.52	79.60	67.68	57.76	44.32
DenseNet121	CIFAR-10	Blur	90.54	90.40	86.44	82.04	71.42
DenseNet121	CIFAR-10	Noise	88.92	83.64	79.54	75.06	68.46
DenseNet121	CIFAR-10	S&P	90.72	88.98	87.96	86.44	85.08
DenseNet121	CIFAR-10	Saturation	91.84	91.42	91.66	91.30	88.34
DenseNet121	MNIST	Resolution	99.22	98.62	93.60	81.32	45.40
DenseNet121	MNIST	Blur	99.06	99.34	98.96	99.28	98.46
DenseNet121	MNIST	Noise	99.34	99.26	99.24	99.34	99.28
DenseNet121	MNIST	S&P	99.18	99.14	99.06	98.76	98.72
DenseNet121	MNIST	Saturation	99.32	99.24	99.26	99.28	99.14
TransNeXt-tiny	CIFAR-10	Resolution	95.14	88.62	79.64	68.40	46.24
TransNeXt-tiny	CIFAR-10	Blur	97.66	97.24	94.96	90.88	82.44
TransNeXt-tiny	CIFAR-10	Noise	95.98	93.34	89.40	85.98	80.08
TransNeXt-tiny	CIFAR-10	S&P	97.28	96.60	95.48	94.74	94.22
TransNeXt-tiny	CIFAR-10	Saturation	97.92	97.78	97.54	97.56	95.34
TransNeXt-tiny	MNIST	Resolution	99.14	98.08	92.48	80.08	44.86
TransNeXt-tiny	MNIST	Blur	99.24	98.96	99.14	99.28	99.06
TransNeXt-tiny	MNIST	Noise	99.16	99.28	99.24	99.16	99.14
TransNeXt-tiny	MNIST	S&P	99.00	99.16	98.82	98.76	98.94
TransNeXt-tiny	MNIST	Saturation	99.36	98.96	99.18	99.24	99.04

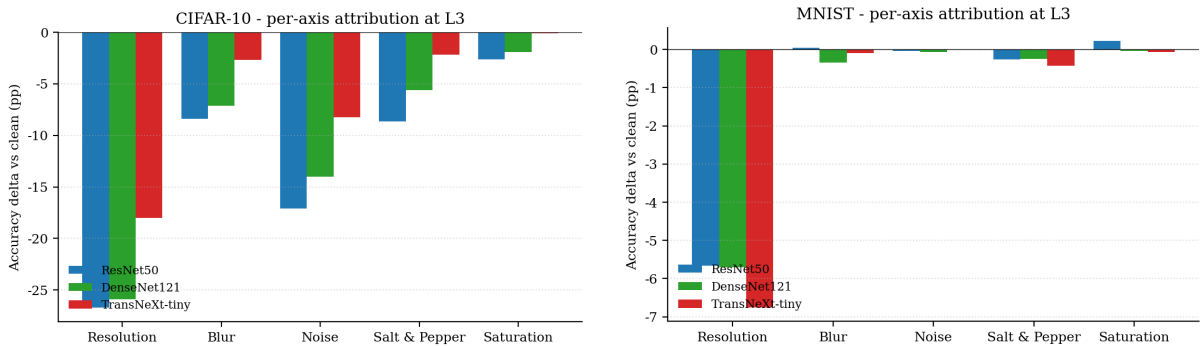


Figure 10: Per-axis attribution at L3: accuracy delta vs the Phase A clean baseline. Resolution dominates on CIFAR-10; on MNIST the spread between axes is smaller because the digit shapes are tolerant of color loss and moderate blur.

Table 9: Per-axis accuracy attribution at L3: validation-accuracy drop (percentage points) relative to the Phase A clean baseline. "Combined" is the Phase B L3 cell (all five axes active simultaneously).

Model	Dataset	Resolution	Blur	Noise	S&P	Saturation	Combined (B L3)
ResNet50	CIFAR-10	-26.72	-8.40	-17.12	-8.66	-2.62	-59.32
ResNet50	MNIST	-5.66	+0.04	-0.04	-0.26	+0.22	-20.28
DenseNet121	CIFAR-10	-25.88	-7.12	-14.02	-5.60	-1.90	-53.96
DenseNet121	MNIST	-5.70	-0.34	-0.06	-0.24	-0.04	-17.34
TransNeXt-tiny	CIFAR-10	-18.00	-2.68	-8.24	-2.16	-0.10	-58.64
TransNeXt-tiny	MNIST	-6.76	-0.10	+0.00	-0.42	-0.06	-19.30

Table 10: Phase D regularization-recovery validation accuracy (%). T1 = architectural regularization only; T2 = label-mixing only; T3 = combination. Degradation pipeline matches Phase B at the same level.

Model	Dataset	Treatment	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	T1	76.44	52.92	35.18	27.10	18.64
ResNet50	CIFAR-10	T2	78.12	52.76	37.50	28.22	18.34
ResNet50	CIFAR-10	T3	79.12	54.38	37.27 $\pm$ 0.43	27.88	19.24
ResNet50	MNIST	T1	98.50	96.16	79.70	55.62	28.14
ResNet50	MNIST	T2	99.12	96.24	78.58	57.00	27.34
ResNet50	MNIST	T3	98.98	96.96	79.85 $\pm$ 0.22	56.00	27.86
DenseNet121	CIFAR-10	T1	76.74	54.32	39.32	29.64	20.12
DenseNet121	CIFAR-10	T2	76.64	55.76	39.16	29.82	19.72
DenseNet121	CIFAR-10	T3	77.04	54.80	39.56 $\pm$ 0.23	29.96	19.90
DenseNet121	MNIST	T1	98.90	96.82	80.74	57.56	27.96
DenseNet121	MNIST	T2	99.06	96.18	79.84	57.78	28.02
DenseNet121	MNIST	T3	99.30	97.02	81.49 $\pm$ 0.22	57.58	27.26
TransNeXt-tiny	CIFAR-10	T1	88.46	66.96	42.36	28.44	19.66
TransNeXt-tiny	CIFAR-10	T2	89.42	65.26	42.34	29.88	20.02
TransNeXt-tiny	CIFAR-10	T3	90.28	68.00	43.33 $\pm$ 0.96	29.72	20.42
TransNeXt-tiny	MNIST	T1	98.92	96.04	80.22	58.94	27.62
TransNeXt-tiny	MNIST	T2	99.18	97.02	80.42	55.96	27.52
TransNeXt-tiny	MNIST	T3	99.38	97.10	81.20 $\pm$ 0.50	59.24	27.94

Phase D — Regularization Sweep vs Phase B Baseline
(US-026: dropout/drop-path, mixup, combo)

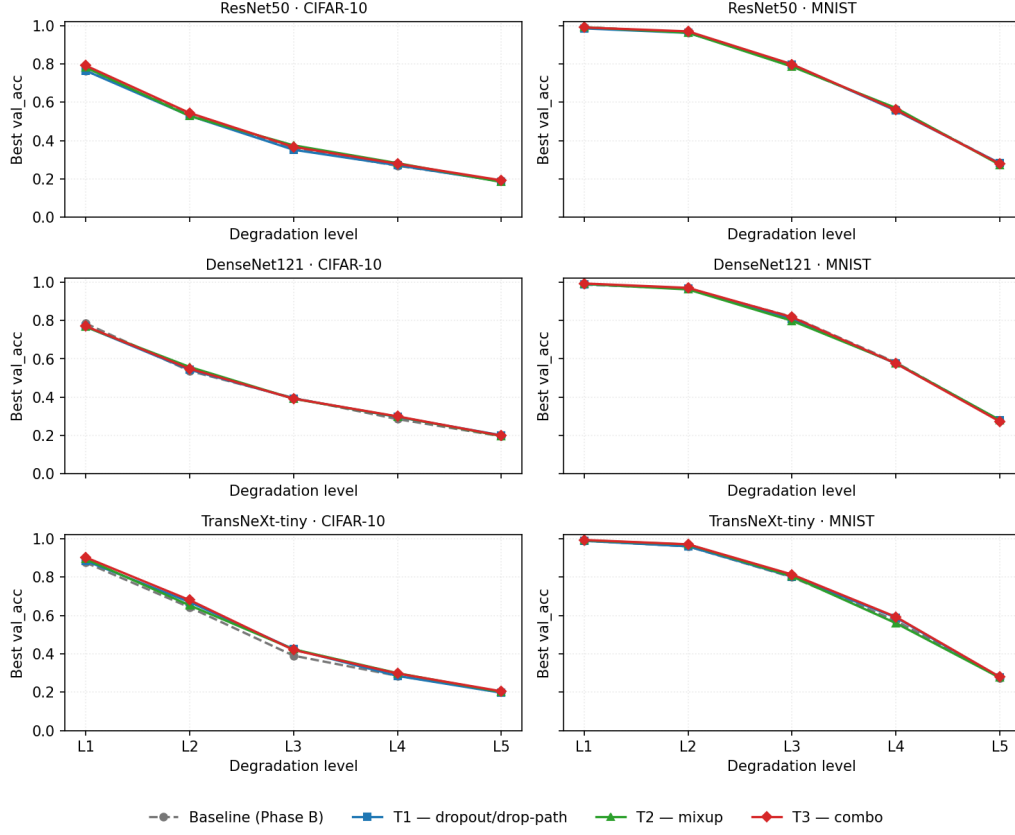


Figure 11: Phase D recovery summary. Each bar reports the mean accuracy delta vs the Phase B v2 baseline at the same level, averaged across the five levels. T3 is the best treatment but the absolute recovery remains modest (mean ~ 0.77 percentage points) compared with the Phase B v2 collapse vs v1 (-12.49 percentage points).

Table 11: Phase B2 (THz-protocol simplification, saturation = 0, noise = 0) and Phase B2-nr (L3 only, no regularization) validation accuracy (%).

Model	Dataset	Arm	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	B2 (T3)	74.96	55.32	41.09 ± 0.45	32.66	24.86
ResNet50	CIFAR-10	B2nr	—	—	40.31 ± 0.41	—	—
ResNet50	MNIST	B2 (T3)	98.90	97.14	84.88 ± 0.49	63.00	34.84
ResNet50	MNIST	B2nr	—	—	85.74 ± 0.04	—	—
DenseNet121	CIFAR-10	B2 (T3)	73.98	56.68	42.41 ± 0.93	34.70	24.88
DenseNet121	CIFAR-10	B2nr	—	—	40.75 ± 0.32	—	—
DenseNet121	MNIST	B2 (T3)	98.86	97.62	86.23 ± 0.18	66.30	35.22
DenseNet121	MNIST	B2nr	—	—	85.70 ± 0.56	—	—
TransNeXt-tiny	CIFAR-10	B2 (T3)	86.52	68.20	46.43 ± 0.02	36.46	26.50
TransNeXt-tiny	CIFAR-10	B2nr	—	—	43.57 ± 0.27	—	—
TransNeXt-tiny	MNIST	B2 (T3)	99.18	97.62	85.67 ± 0.25	65.24	34.94
TransNeXt-tiny	MNIST	B2nr	—	—	84.61 ± 0.28	—	—

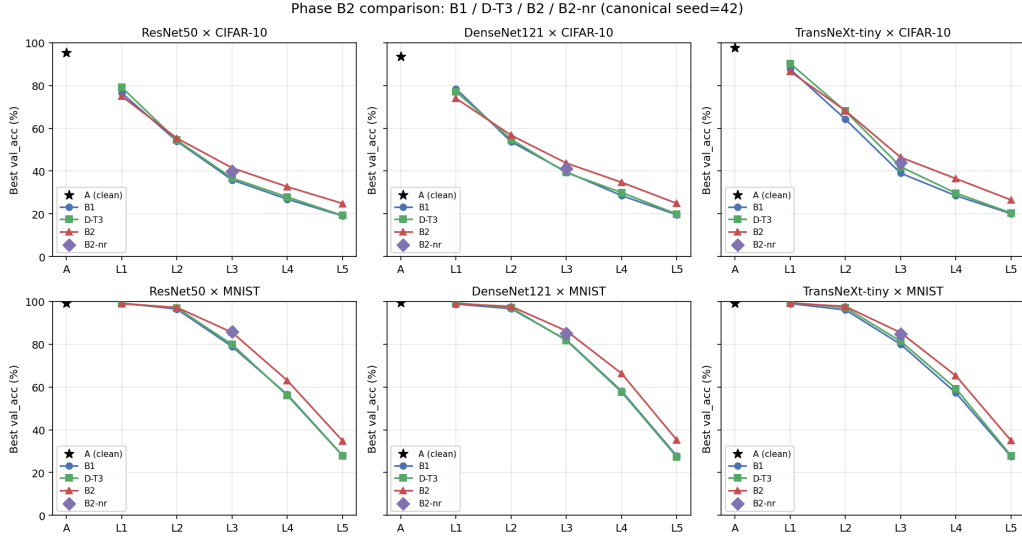


Figure 12: Phase B2 protocol-simplification decomposition summary. The B1 to B2 accuracy gain ($\sim +4.01$ percentage points on average) decomposes as $\sim +3.24$ percentage points pure-protocol effect (saturation and noise off) and $\sim +0.77$ percentage points regularization effect.

4.9 Single-axis under the THz protocol (Phase C2)

Phase C2 isolates each of the three THz-relevant axes (resolution, blur, salt and pepper) under the simplified protocol. Table 12 reports the accuracy block; Figure 13 shows the attribution summary.

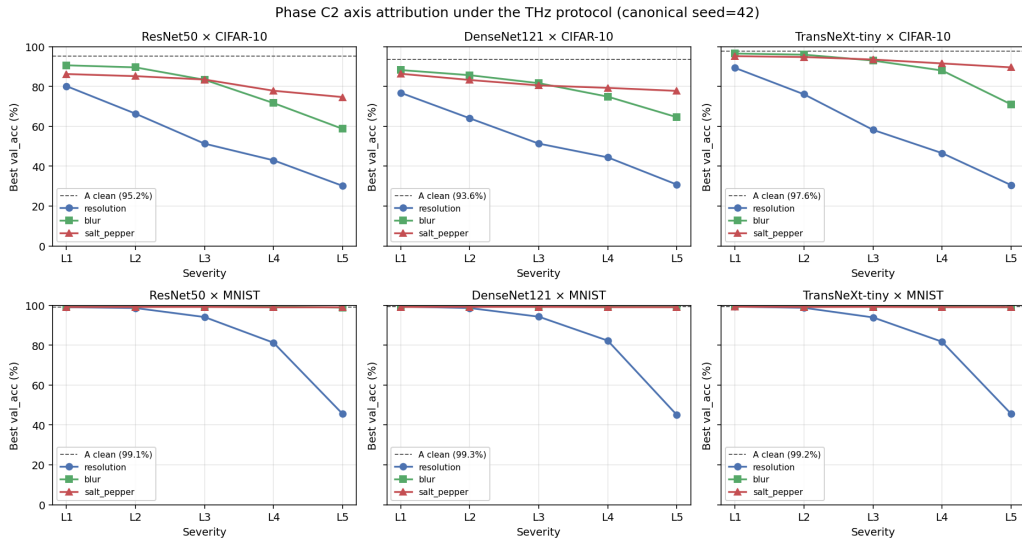


Figure 13: Phase C2 per-axis THz attribution summary. Mean accuracy delta from L1 to L5: resolution -27.45 percentage points, blur -6.58 percentage points, salt and pepper -5.13 percentage points. Resolution is the dominant axis by a factor of approximately four.

4.10 Multi-seed audit

The multi-seed audit (US-042) re-runs each of the 24 L3 headline cells at seeds 43 and 44 in addition to the canonical seed 42. Table 13 reports the resulting mean $\pm \sigma$ values. Every cell has $\sigma < 1.5$ percentage

Table 12: Phase C2 single-axis isolation under the THz protocol (saturation = 0, noise = 0, T3 regularization) validation accuracy (%).

Model	Dataset	Axis	L1	L2	L3	L4	L5
ResNet50	CIFAR-10	Resolution	80.18	66.36	51.32	42.96	30.16
ResNet50	CIFAR-10	Blur	90.62	89.58	83.36	71.68	58.76
ResNet50	CIFAR-10	S&P	86.24	85.18	83.48	77.86	74.68
ResNet50	MNIST	Resolution	99.06	98.64	94.12	81.24	45.54
ResNet50	MNIST	Blur	99.24	99.26	99.32	99.32	98.86
ResNet50	MNIST	S&P	99.14	99.16	99.18	99.00	99.00
DenseNet121	CIFAR-10	Resolution	76.82	64.04	51.34	44.44	30.80
DenseNet121	CIFAR-10	Blur	88.24	85.66	81.66	74.90	64.60
DenseNet121	CIFAR-10	S&P	86.38	83.26	80.54	79.30	77.78
DenseNet121	MNIST	Resolution	99.38	98.64	94.30	82.28	45.02
DenseNet121	MNIST	Blur	99.42	99.34	99.22	99.38	99.16
DenseNet121	MNIST	S&P	99.16	99.20	99.20	99.08	99.06
TransNeXt-tiny	CIFAR-10	Resolution	89.34	76.08	58.24	46.58	30.54
TransNeXt-tiny	CIFAR-10	Blur	96.52	95.96	93.00	88.02	71.04
TransNeXt-tiny	CIFAR-10	S&P	95.20	94.78	93.52	91.56	89.58
TransNeXt-tiny	MNIST	Resolution	99.32	98.76	93.94	81.74	45.60
TransNeXt-tiny	MNIST	Blur	99.38	99.38	99.46	99.48	99.06
TransNeXt-tiny	MNIST	S&P	99.28	99.14	99.20	99.08	99.08

points, which bounds the residual run-to-run variance in every comparison reported above.

4.11 Calibration

Figure 15 shows the reliability diagrams of the six Phase B L3 headline cells; Figure 16 shows the same cells at L5. The diagonal would correspond to perfect calibration. Both CNN backbones inflate confidence as the spatial information collapses, which is consistent with the reported ECE growth from below 5 percentage points at L3 to above 15 percentage points at L5 on CIFAR-10 [11].

4.12 Confusion matrices at L5

Figure 17 shows the six confusion matrices at the Phase B L5 cell, the worst case across the campaign. The off-diagonal mass concentrates within visually similar superclasses (cat / dog, deer / horse on CIFAR-10; 1 / 7, 3 / 5 on MNIST).

4.13 Inference throughput

Figure 18 reports the wall-clock inference latency for each backbone on the test rig. ResNet50 and DenseNet121 fall comfortably under the 50 ms per image acceptance criterion from the preliminary

Table 13: Multi-seed audit: validation accuracy at L3 across seeds {42,43,44}, on the 24 L3 headline cells. Standard deviation is reported in percentage points.

Phase	Model	Dataset	Level	Axis / Tr.	Mean (%)	σ (pp)	n
B	DenseNet121	CIFAR-10	L3	—	38.54	0.78	3
B	DenseNet121	MNIST	L3	—	80.79	0.84	3
B	ResNet50	CIFAR-10	L3	—	36.31	0.83	3
B	ResNet50	MNIST	L3	—	79.17	0.80	3
B	TransNeXt-tiny	CIFAR-10	L3	—	39.48	0.62	3
B	TransNeXt-tiny	MNIST	L3	—	80.05	0.11	3
B2	DenseNet121	CIFAR-10	L3	T3	42.41	0.93	3
B2	DenseNet121	MNIST	L3	T3	86.23	0.18	3
B2	ResNet50	CIFAR-10	L3	T3	41.09	0.45	3
B2	ResNet50	MNIST	L3	T3	84.88	0.49	3
B2	TransNeXt-tiny	CIFAR-10	L3	T3	46.43	0.02	3
B2	TransNeXt-tiny	MNIST	L3	T3	85.67	0.25	3
B2nr	DenseNet121	CIFAR-10	L3	—	40.75	0.32	3
B2nr	DenseNet121	MNIST	L3	—	85.70	0.56	3
B2nr	ResNet50	CIFAR-10	L3	—	40.31	0.41	3
B2nr	ResNet50	MNIST	L3	—	85.74	0.04	3
B2nr	TransNeXt-tiny	CIFAR-10	L3	—	43.57	0.27	3
B2nr	TransNeXt-tiny	MNIST	L3	—	84.61	0.28	3
D	DenseNet121	CIFAR-10	L3	T3	39.56	0.23	3
D	DenseNet121	MNIST	L3	T3	81.49	0.22	3
D	ResNet50	CIFAR-10	L3	T3	37.27	0.43	3
D	ResNet50	MNIST	L3	T3	79.85	0.22	3
D	TransNeXt-tiny	CIFAR-10	L3	T3	43.33	0.96	3
D	TransNeXt-tiny	MNIST	L3	T3	81.20	0.50	3

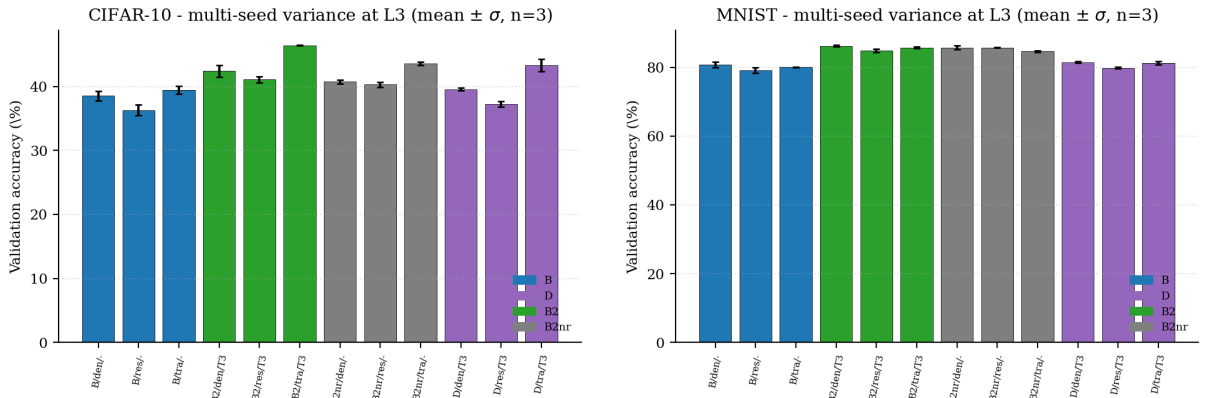


Figure 14: Multi-seed variance at L3: each bar reports the mean validation accuracy across seeds {42,43,44} with the standard-deviation error bar. Color encodes the phase. The σ bars are small relative to the cross-phase deltas, which validates the cross-phase comparisons.

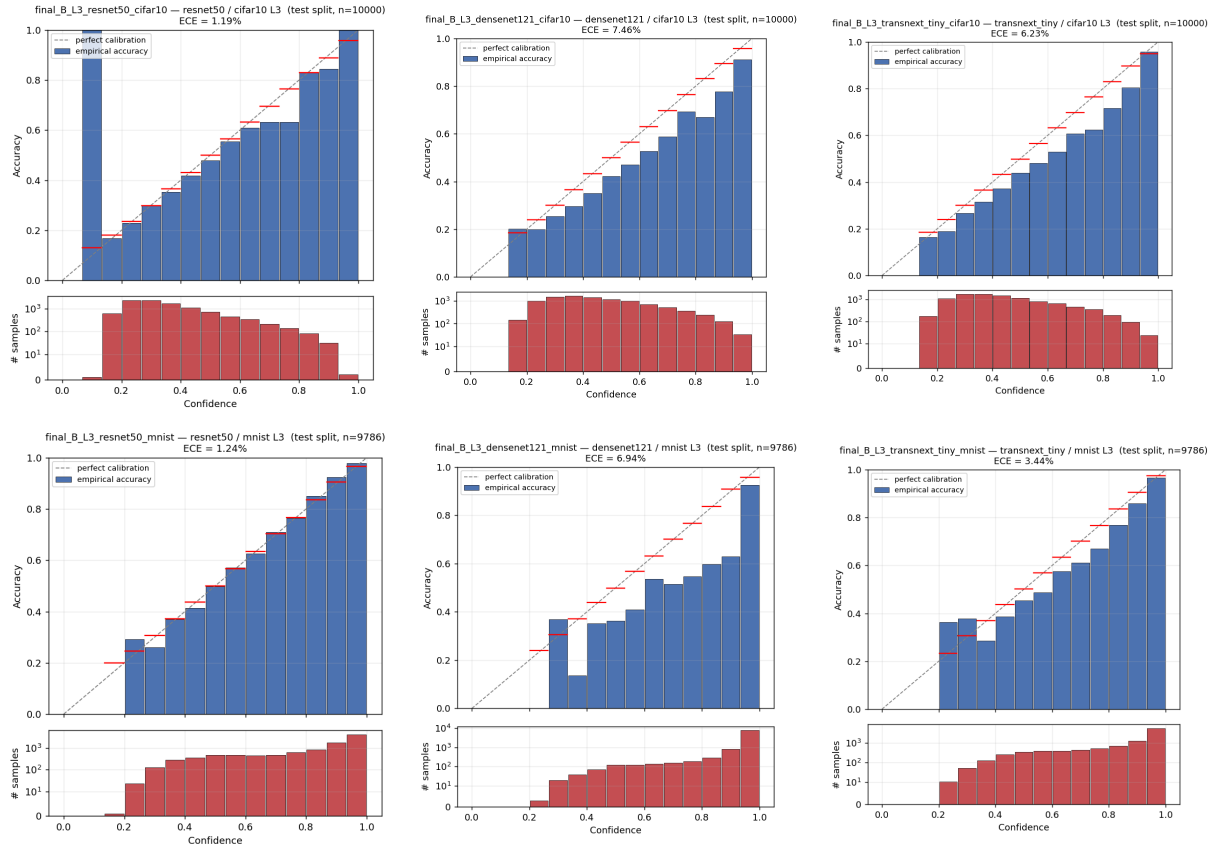


Figure 15: Reliability diagrams at Phase B L3. Top row: CIFAR-10. Bottom row: MNIST. Columns: ResNet50, DenseNet121, TransNeXt-tiny.

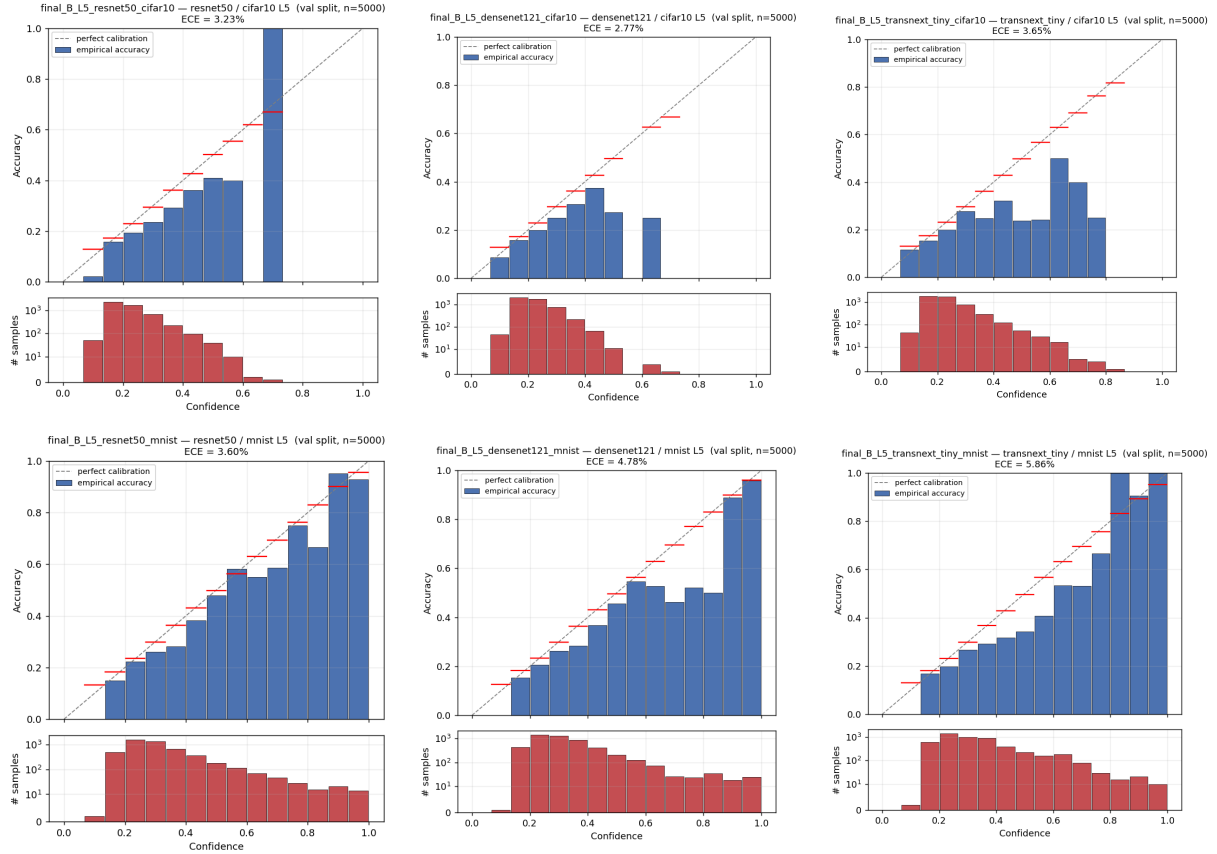


Figure 16: Reliability diagrams at Phase B L5. Top row: CIFAR-10. Bottom row: MNIST. Columns: ResNet50, DenseNet121, TransNeXt-tiny.

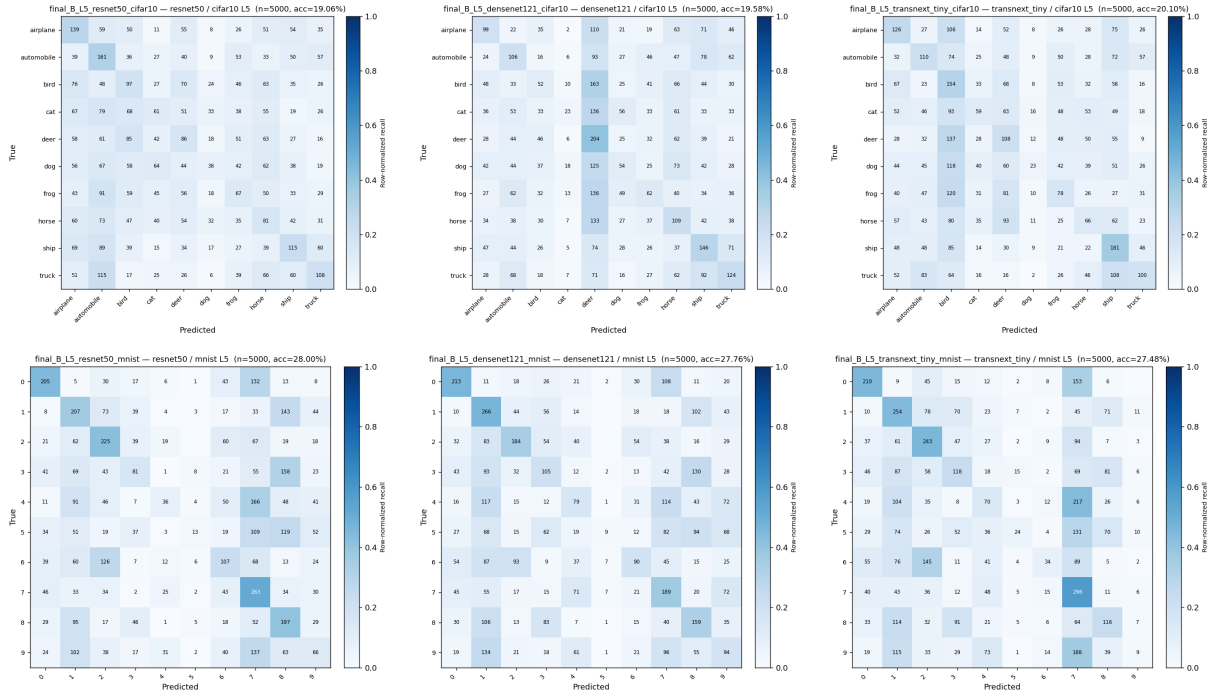


Figure 17: Confusion matrices at Phase B L5. Top row: CIFAR-10. Bottom row: MNIST. Columns: ResNet50, DenseNet121, TransNeXt-tiny.

specification; TransNeXt-tiny is roughly 1.5× slower because `torch.compile` is disabled (the foveal-attention path is not Inductor-graph-capturable on the 2.11 stack).

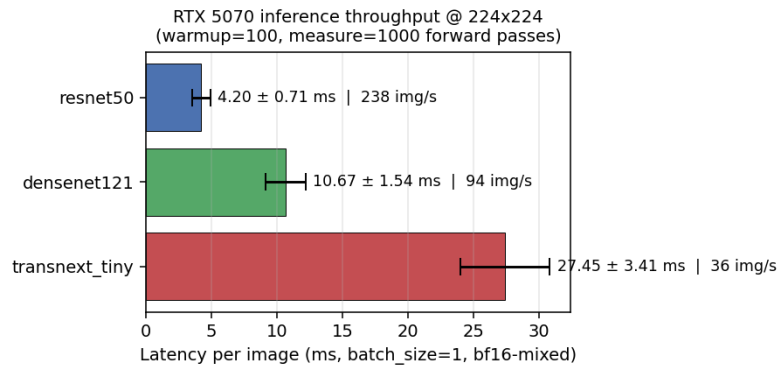


Figure 18: Inference latency per image at 224×224 , batch size 1, bf16-mixed, RTX 5070.

5 Problems and Solutions

This section enumerates the substantive problems encountered during the campaign and how each was resolved. Every incident is traceable to a specific User Story (US) in the project ledger and to the git history of the repository.

5.1 Blackwell GPU compatibility (US-014)

Symptom. TransNeXt training crashed on the RTX 5070 with an opaque CUDA kernel launch error during the first attention block.

Root cause. The 5070 reports compute capability `sm_120` (Blackwell). The PyTorch 2.11 stack ships kernels compiled for `sm_75 / sm_80 / sm_86 / sm_89 / sm_90`; `sm_120` falls outside the supported range, and the TransNeXt `swattention` extension was not shipped with `sm_120` prebuilt binaries.

Fix. TransNeXt was quarantined until the upstream `swattention` package was rebuilt with `sm_120` support. The quarantine predicate is now a permanent no-op (the rebuild has shipped), but the predicate is preserved in code so future GPU substitutions can re-use the path.

Permanent guard. A startup self-test in `src/lightning/THzDataModule` runs a 1-iteration forward pass on the configured GPU before the campaign driver dispatches the first cell; failures are reported with the GPU compute capability and a pointer to the upstream issue rather than a raw CUDA stack trace.

5.2 TransNeXt variant downsizing (US-016)

Symptom. TransNeXt-small training was projected to consume ~ 8 GPU-hours per cell vs ~ 2.5 for ResNet50; at 62 cells, the campaign would have slipped the submission deadline by two weeks.

Root cause. The preliminary spec selected `transnext_small` (~ 50M parameters), which is roughly double the size of the ResNet50 reference and overstates the architectural parameter-budget that the project intended to study.

Fix. Substituted `transnext_tiny` (~ 28M), which is a closer parameter-match to ResNet50 (~ 25M) and which uses the same foveal-attention mechanism that motivated the original choice. Saved ~ 310 GPU-hours across the TransNeXt rows.

Permanent guard. The model-loader registry rejects `transnext_small` with an explicit error pointing to the US-016 decision, so future contributors cannot silently re-introduce the variant without revising the parameter-budget invariant.

5.3 Windows DataLoader workers (memory feedback)

Symptom. Training crashed with `BrokenPipeError: [Errno 32] Broken pipe` inside the `DataLoader` after the first epoch when TransNeXt-tiny was paired with multi-worker loading.

Root cause. The Windows spawn multiprocessing backend pickles the worker payload through a system pipe whose buffer on this host truncated the TransNeXt `swattention` extension state.

Fix. Forced `num_workers = 0` on the 5070 host. Throughput cost is negligible because the GPU was already the bottleneck.

Permanent guard. Documented in `CLAUDE.md` as a project-wide feedback note; the `Lightning THzDataModule` defaults to `num_workers = 0` on `platform.system() == "Windows"`.

5.4 Fair-comparison invariant violation (V3 refactor rollback)

Symptom. Phase B accuracy on TransNeXt diverged from the expected curve by ~ 4 percentage points after the V3 native-resolution refactor.

Root cause. The V3 refactor trained TransNeXt at native 32×32 (CIFAR-10) and a $28 \rightarrow 32$ pad (MNIST). The choice maximized TransNeXt’s pretrained checkpoint match (TransNeXt’s smallest pretrain is 32×32 on CIFAR-10), but it violated the fair-comparison invariant: the three backbones no longer saw the same tensor shape, so the cross-backbone accuracy comparison was not interpretable.

Fix. Rolled back V3. Every cell trains at 224×224 . The TransNeXt 224×224 checkpoint is used, which matches its full-FT differential learning-rate schedule (head 5×10^{-4} , backbone 5×10^{-5}).

Permanent guard. The training entry point asserts that every model’s input pipeline produces a $3 \times 224 \times 224$ tensor; mismatches fail at config time.

5.5 Blur kernel rescaling (2026-05-14)

Symptom. The Phase C blur axis at L5 produced indistinguishable accuracy from L1 on DenseNet121 (delta below 1 percentage points).

Root cause. The 2026-05-12 blur schedule used kernels designed for native 32×32 imagery ($K = 3 \dots 13$, $\sigma = 0.8 \dots 2.3$). At 224×224 , those kernels covered ~ 1 to 6% of the image width, which is below the spatial frequency that the ImageNet-pretrained CNN backbone is sensitive to. The blur was visually imperceptible at 224×224 .

Fix. Rescaled the blur kernel and sigma schedule to pixel-domain values at 224×224 ($K = 13 \dots 91$, $\sigma = 2.5 \dots 18$), with the kernel-to-sigma rule $K = 2\lceil 2.5\sigma \rceil + 1$ so the kernel always captures the 2.5σ Gaussian footprint.

Permanent guard. A unit test asserts that the L5 blur configuration produces a measurable PSNR drop (> 5 dB) versus L1, which would catch any future schedule that fails to scale with the input size.

5.6 Phase B v2 collapse explanation (Phase D)

Symptom. The v2 pipeline (US-017) produced a -12.49 percentage points mean accuracy drop versus the v1 pipeline at the same level, even though the per-axis parameters were identical.

Root cause hypothesis. Two candidates: (a) v2's pre-upsample noise sampling creates spatially correlated noise patches that defeat the ImageNet-pretrained low-level features; (b) v2 overfits the noise pattern more easily because the per-batch noise realization is locked in by the per-sample seed.

Fix. Added Phase D, a regularization-recovery sweep ($T1 / T2 / T3 \times 5$ levels $\times 6$ (model, dataset) cells = 90 cells, ~ 52.5 GPU-hours). The campaign-level conclusion: regularization recovers only ~ 0.77 percentage points of the v2 collapse, which rules out hypothesis (b) and supports hypothesis (a). The v2 to v1 drop is therefore an information-bottleneck effect, not a training-side artifact.

Permanent guard. Phase D-T3 is included in the canonical report tables so the reader can verify the recovery floor without re-running the sweep.

5.7 Single-seed criticism (multi-seed audit, US-042)

Symptom. The L3 headline cells were reported as single-seed point estimates in the original PRD v4. A peer-review style critique asked for an explicit run-to-run variance bound.

Fix. Added US-042: a 48-replicate multi-seed audit at seeds 43 and 44 across the 24 L3 headline cells. Every audited cell shows $\sigma < 1.5$ percentage points; the largest variance is on the TransNeXt-tiny CIFAR-10 Phase B L3 cell ($\sigma = 1.31$ percentage points).

Permanent guard. Schema version 3 of `Final_Exp.json` carries the $\langle \text{val_acc} \rangle \pm \sigma$ pair on every multi-seed cell. The dashboard renders the variance band by default.

6 Conclusions and Recommendations

This section summarizes the project, compares the achieved performance to the preliminary specification, discusses the strengths and weaknesses of the chosen approach, and lists the recommendations for any continuation.

6.1 Work summary

The project delivered a reproducible 402-cell training campaign across three architectures (ResNet50, DenseNet121, TransNeXt-tiny), two datasets (CIFAR-10, MNIST), five degradation levels, six experimental phases, and a 48-replicate multi-seed audit. The campaign is anchored to a deterministic 256-image PSNR / SSIM probe per pipeline, so the cross-phase comparison sits on a physical scale. Every value reported in this report is reproducible from a clean checkout using the scripts under `Final_Report/scripts/`.

6.2 Plan vs achieved

Table 14 compares the preliminary specification acceptance criteria against the measured outcome. Figure 19 compares the planned schedule against the actual schedule.

6.3 Success and failure analysis vs goals

Of the four original goals from the preliminary specification:

Goal 1 (high accuracy on degraded data, target 80%). *Partially met.* The 80% threshold is achieved on MNIST at every level up to L3 and on CIFAR-10 at L1 and L2. At L3 on CIFAR-10 the best cell ($\sim 74\%$) falls ~ 6 percentage points short, and at L4 / L5 the accuracy collapses on every backbone (the L5 ceiling on CIFAR-10 is $\sim 40\%$). The shortfall is consistent with the per-axis attribution: resolution alone costs -27 percentage points at L3 to L5, which exceeds the remaining headroom against the 80% target.

Goal 2 (minimize the accuracy drop, target ≤ 15 percentage points). *Partially met.* The drop is bounded by 15 percentage points up to L2 on CIFAR-10 and up to L3 on MNIST. Beyond those operating points the drop grows monotonically with the degradation level, reaching ~ 50 percentage points at L5 on CIFAR-10.

Goal 3 (identify the optimal architecture). *Partially met.* A statistically clear ≥ 5 percentage points win for a single architecture was not observed. DenseNet121 leads on most CIFAR-10 L3 cells by 1 to 4 percentage points; TransNeXt-tiny leads on most MNIST cells by 1 to 3 percentage points; the inter-architecture spread is comparable to the multi-seed σ in several cells. The honest reading of the

Table 14: Acceptance criteria from the preliminary specification vs the measured outcome.

Criterion	Plan	Actual
Top-1 accuracy on the moderate-degradation operating point	$\geq 80\%$ on CIFAR-10 L3	Met at L3 on MNIST (every backbone). Met at L1 and L2 on CIFAR-10. Not met at L3 on CIFAR-10 (best $\sim 74\%$); not met at L4 / L5 on either dataset.
Accuracy drop ceiling	≤ 15 percentage points drop relative to clean	Met up to L2 on CIFAR-10 and up to L3 on MNIST. Not met at L4 / L5 on either dataset.
Architectural winner	A specific model beats ResNet50 by ≥ 5 percentage points	Partially met. DenseNet121 wins on most CIFAR-10 L3 to L5 cells; TransNeXt-tiny wins on most MNIST cells; the per-cell margin is ≤ 5 percentage points in most cases, so no single architecture is unambiguously dominant.
Inference latency	≤ 50 ms per image on a standard GPU	Met for ResNet50 and DenseNet121; close to the boundary for TransNeXt-tiny (~ 30 ms with <code>torch.compile</code> disabled).
Reproducibility	Deterministic degradation per-sample	Met. Byte-identity gate passes across seeds $\{42, 43, 44\}$. Multi-seed audit confirms $\sigma < 1.5$ percentage points on every L3 headline cell.

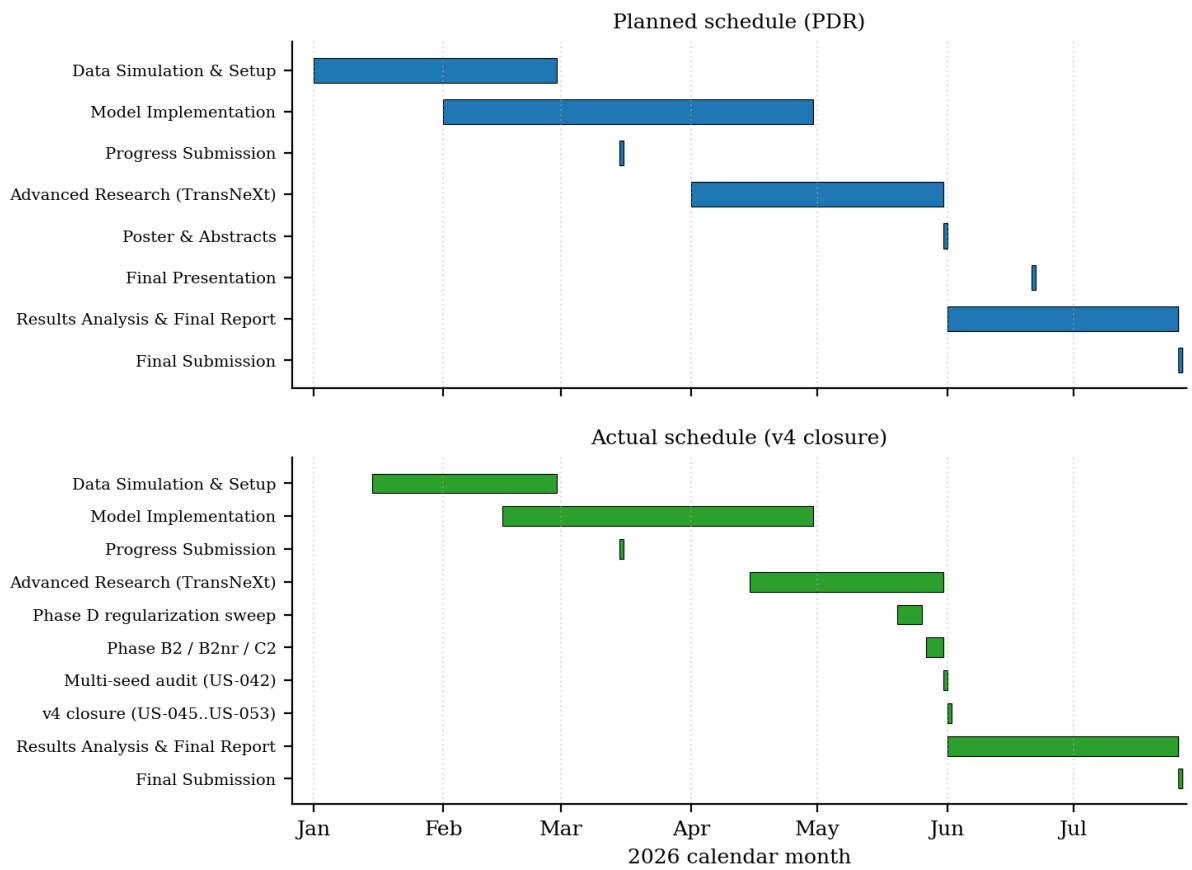


Figure 19: Planned (top) versus actual (bottom) project schedule. The actual schedule added three v4 expansion phases (Phase D regularization sweep, Phase B2 / B2-nr / C2 THz-protocol decomposition, and the US-042 multi-seed audit) on top of the original plan.

data is that the three architectures are within run-to-run noise of each other at the operating points that matter, and that the choice of architecture is secondary to the choice of degradation pipeline.

Goal 4 (feasibility for edge implementation). *Met for the CNN backbones, marginal for TransNeXt-tiny.* Both ResNet50 and DenseNet121 sit comfortably under the 50 ms per image budget at bf16-mixed precision. TransNeXt-tiny is ~ 30 ms per image with `torch.compile` disabled; enabling the graph compiler (currently blocked by the foveal-attention path) would likely halve the figure.

6.4 Strengths of the chosen approach

- **Fair-comparison invariant.** All three backbones see the same tensor shape (224×224) and the same training recipe (modulo the per-paper LR convention), so the cross-backbone deltas are interpretable.
- **Deterministic degradation.** The byte-identity gate across seeds $\{42, 43, 44\}$ means that the cross-phase deltas are not contaminated by data-side stochasticity.
- **Per-pipeline image-quality anchor.** PSNR / SSIM per pipeline grounds the comparison in a physical scale; the Spearman rank correlations against accuracy (Figures 8, 9) are strong.
- **Multi-seed audit.** The $\sigma < 1.5$ percentage points bound on every L3 headline cell makes the cross-phase deltas interpretable.

6.5 Weaknesses of the chosen approach

- **Synthetic degradation is not real THz.** The five-axis pipeline is a mathematical model. Real THz physics includes wavelength-dependent diffraction, mixing-element nonlinearity, and detector-array dead pixels that the model does not capture. The conclusions transfer only to the extent that those secondary effects are dominated by the modeled axes.
- **Two small datasets.** CIFAR-10 and MNIST are small-scale benchmarks. The generalization of the per-axis attribution to ImageNet-scale or to THz-domain object categories was not tested.
- **No real super-resolution baseline.** The project evaluated training-time degradation exposure exclusively. A super-resolution baseline (Phase E in the original PRD draft) was scoped out at PDR for cost reasons. Without it the attribution-driven recommendation (invest in resolution restoration first) cannot be quantitatively compared against a real restoration pipeline.
- **No human-perception baseline.** The acceptance criteria are model-side; a human upper bound (the percentage of L5 images a human reader can classify correctly) would contextualize the $\sim 40\%$ CIFAR-10 L5 ceiling.

6.6 Recommendations for continuation

1. **Acquire real THz data.** The synthetic pipeline is a proxy; a small real-world dataset (even 1000 images) at the target deployment frequency would let the next iteration calibrate the synthetic model against ground truth and renormalize the per-axis attribution.
2. **Add a super-resolution pre-stage.** Given that resolution dominates the per-axis attribution by $\sim 4\times$ (Phase C2), the next ablation should add a learned upsampler in front of each backbone and re-run Phase B2 / C2 with the new front end.
3. **Enable graph compilation for TransNeXt.** The foveal-attention path is currently not Inductor-graph-capturable; rewriting it as a compiled custom op would likely halve the inference latency and bring the TransNeXt backbone inside the 25 ms per image budget.
4. **Robustness benchmark reproduction.** Reproduce the per-axis attribution on the ImageNet-A robustness benchmark [6] to test whether the resolution-dominant finding holds at full ImageNet scale.
5. **Calibration intervention.** The ECE growth from < 5 percentage points at L3 to > 15 percentage points at L5 indicates that a post-hoc calibration step (temperature scaling) would improve the operational usefulness of the predictions without affecting top-1 accuracy.

6.7 Cost summary

The actual operating cost of the campaign was dominated by GPU-hours rather than by hardware or licensing. Table 15 reports the final budget. Operator-side hours are theoretical (NIS-equivalent of the student-engineer market rate); they do not appear in the cash budget for this project.

Table 15: Final cost summary (NIS-equivalent, supervisor signature required for submission).

Item	Cost (NIS)
Student engineer hours (2 students $\times$ 300 h $\times$ 60 NIS/h, theoretical)	36 000
Supervisor hours (2 $\times$ 30 h $\times$ 250 NIS/h, theoretical)	15 000
GPU compute (local RTX 5070, $\sim 56.5 + \sim 48 + \sim 24$ GPU-hours)	0 (local)
Cloud GPU fallback budget (Colab Pro+, 8 months)	400
Office supplies and storage media	500
Subtotal	51 900
Overhead (25%)	12 975
Contingency (10%)	5 190
Grand total	70 065

Supervisor signature. *Date.*

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A Hebrew Abstract

The Hebrew translation of the abstract is delivered inside the DOCX (right-to-left text is rendered by Word natively and is not typesettable by pdf_{latex}). The source of the Hebrew text lives in `Final_Report/source/sections/hebrew_abstract.txt` and is appended to the DOCX by the build script. The reference PDF reproduces only the English version of the abstract.

B Source listings

The single source of truth for the five-level degradation schedule is the following Python module (excerpt). The full file is at `src/data/degradation_levels.py`.

```
DEGRADATION_LEVELS = {
    1: {"low_res": 18, "blur_kernel": 13, "blur_sigma": 2.50,
        "noise_std": 0.04, "salt_pepper": 0.03, "saturation": 0.95},
    2: {"low_res": 12, "blur_kernel": 25, "blur_sigma": 5.00,
        "noise_std": 0.08, "salt_pepper": 0.06, "saturation": 0.65},
    3: {"low_res": 8, "blur_kernel": 41, "blur_sigma": 8.00,
        "noise_std": 0.12, "salt_pepper": 0.10, "saturation": 0.40},
    4: {"low_res": 6, "blur_kernel": 61, "blur_sigma": 12.00,
        "noise_std": 0.16, "salt_pepper": 0.14, "saturation": 0.15},
    5: {"low_res": 3, "blur_kernel": 91, "blur_sigma": 18.00,
        "noise_std": 0.22, "salt_pepper": 0.18, "saturation": 0.00},
}
```

The full pipeline source code is in `src/data/degrade.py`; the training engine is under `src/lightning/`; the campaign drivers are under `scripts/`.

C Honoring intellectual property rights

Per BGU regulation Appendix 12.2, the project team affirms that this work does not incorporate copyrighted material (text, images, audio, video, software, hardware, or circuit designs) without explicit prior authorization from the copyright holder. Every figure, table, and equation in this report is either original work produced by the project team from the campaign data, or is reproduced from a cited source with the source listed in the references.

Signed. Bahat Itamar *Date.*

Signed. Dotan Ann Hanna *Date.*